



**TRIBHUVAN UNIVERSITY
ASIAN SCHOOL OF MANAGEMENT & TECHNOLOGY**

**Final Year Project Report
On
“Futsal Bookings and Management Platform”**

Submitted To :

Office of Dean
Institute of Science and Technology

**Under the supervision of
Mr. Surya Bam**

*In partial fulfillment of the requirement for the bachelor's degree in Computer
Science and
Information Technology*

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SUPERVISOR’S RECOMMENDATION

It is my pleasure to recommend that a report on “Futsal Bookings and Management Platform” has been prepared under my supervision by Aashish Rijal, Dipesh Devkota, and Kedar Dhungana in partial fulfillment of the requirement of the degree of Bachelor of Science in Computer Science and Information Technology (BSc.CSIT). Their report is satisfactory to process for the future evaluation.

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With respect,

Aashish Rijal

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ABSTRACT

The Futsal Booking and Management System is a web-based multi-tenant SaaS platform designed to digitize and streamline futsal court booking and management in Nepal. Addressing the widespread reliance on manual booking methods, such as phone calls and handwritten registers the system provides a centralized platform where users can search, book, and review futsal centers, while owners manage schedules, pricing, and analytics through a dedicated dashboard. Built using React, Node.js, and MySQL, the platform incorporates machine learning features including Logistic Regression for sentiment analysis, and the Haversine formula for location-based court recommendations. Developed following the Agile methodology, the system aims to eliminate double bookings, improve operational efficiency, and deliver an intelligent, scalable solution for futsal facility management.

Keywords: SaaS, Multi-tenant, Sentiment Analysis, Node Js, React Js, Centralized, Logistic Regression, Agile.

TABLE OF CONTENT

TITLE PAGE	ii
SUPERVISOR'S RECOMMENDATION	iii
CERTIFICATE OF APPROVAL	iv
ACKNOWLEDGEMENT	v
ABSTRACT	vi
TABLE OF CONTENT	vii
LIST OF ABBREVIATIONS	viii
LIST OF FIGURES	ix
LIST OF TABLES	xi
 CHAPTER 1: INTRODUCTION	 1
1.1 Introduction	2
1.2 Problem Statement	2
1.3 Objectives	2
1.4 Scope and Limitation	3
1.5 Development Methodology	4
1.6 Report Organization	5
 CHAPTER 2: LITERATURE REVIEW	 6
2.1 Background Study	6
2.2 Literature Review	6
 CHAPTER 3: SYSTEM ANALYSIS	 9
3.1 System Analysis	9
3.1.1 Requirement Analysis	9
3.1.2 Feasibility Study	16
 CHAPTER 4: SYSTEM DESIGN	 19
4.1 Design Overview	19
4.2 Interface Design	34
4.3 Algorithm Details	34
 CHAPTER 5: IMPLEMENTATION AND TESTING	 38
5.1 Implementation	38
5.2 Testing	41
5.3 Result Analysis	42
 CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATION	 68
6.1 Conclusion	68
6.2 Future Recommendations	69
 REFERENCES	 71
APPENDICES	72
SUPERVISOR'S LOG	73

LIST OF ABBREVIATIONS

API	Application Programming Interface
AWS	Amazon Web Services
DB	Database
HLD	High Level Design
JWT	JSON Web Token
ML	Machine Learning
MySQL	My Structured Query Language
NLP	Natural Language Processing
Oauth	Open Authorization
OTP	One Time Password
PWA	Progressive Web Application
Redis	Remote Dictionary Server
SaaS	Software as a Service
TF-IDF	Term Frequency - Inverse Document Frequency
URL	Uniform Resource Locator
VPS	Virtual Private Server

LIST OF FIGURES

Figure 1.1: Agile Development Model	5
Figure 3.1: Use Case Diagram of Futsal Bookings and management	11
Figure 3.2 : Dataset records	12
Figure 3.3: Gantt Chart	17
Figure 3.4 : Er Diagram 1	18
Figure 3.5 : Er Diagram 2	18
Figure 3.6 : Er Diagram 3	18
Figure 3.7 : DFD level-0 Diagram	19
Figure 3.8 : DFD level-1 Diagram	20
Figure 3.9 : DFD level-2 process 1 Diagram	21
Figure 3.10 : DFD level-2 process 2 Diagram	21
Figure 3.11 : DFD level-2 process 3 Diagram	21
Figure 3.12 : DFD level-2 process 4 Diagram	21
Figure 3.13 : DFD level-2 process 5 Diagram	21
Figure 3.14 : DFD level-2 process 6 Diagram	21
Figure 3.15: Sequence Diagram of Futsal Bookings and Management System	22
Figure 4.1: High-Level System Design of Futsal Bookings and Management System	19
Figure 4.2: Registration and Login Page Design	20
Figure 4.3 : Forum Page Design	21
Figure 4.4: Pricing Page Design	22
Figure 4.5: Futsal listing Page Design	23
Figure 4.6 : Futsal Page Design	24
Figure 4.7 : User Profile page Design	25
Figure 4.8: Contact page Design	44
Figure 5.1 : Test Scenario - User Registration	43
Figure 5.2 : Test Scenario - Set Password	43
Figure 5.3 : Test Scenario - Verify Account	44

Figure 5.4: Test Scenario - User Login	44
Figure 5.5: Test Scenario - Create Forum post	46
Figure 5.6 : Test Scenario - View Forum post	46
Figure 5.7 : Test Scenario- Forum Comment	47
Figure 5.8 : Test Scenario - Empty validation	47
Figure 5.9 : Test Scenario- Payment Initiate	49
Figure 5.10 : Test Scenario- Payment Successful	49
Figure 5.11 : Test Scenario -View available slot	51
Figure 5.12 : Test Scenario- Booking Successful	51
Figure 5.13 : Test Scenario - Booking Notification	52
Figure 5.14 : Test Scenario - Booking History	52
Figure 5.15 : Test Scenario - Listing Futsal	54
Figure 5.16 : Test Scenario - Add pitch	54
Figure 5.17 : Test Scenario -Manage Schedule and Price	55
Figure 5.18 : Test Scenario - Anaytices dashboard	55
Figure 5.19 : Test Scenario - Handle Subscription	56
Figure 5.20 : Test Scenario- Bookings Details	56
Figure 5.21 : Test Scenario - Positive Sentiment Review	58
Figure 5.23 : Test Scenario - Recommendation of Futsal	58
Figure 5.22 : Test Scenario - Negative Sentiment Review	59
Figure 5.24 : Test Scenario - Handle single Review	59
Figure 5.25 Test Scenario - Empty Review Text	60

LIST OF TABLES

Table 3.1 Use Case Scenario for User Operations	12
Table 5.1: Unit Test Cases	42
Table 5.2: Test Cases for Forum Feature	45
Table 5.3: Test Cases for Payment Feature	48
Table 5.4: Test Cases for Booking Feature	50
Table 5.5: Test Cases for Futsal Owner Feature	53
Table 5.6: Test Cases for ML Sentiment Analysis Feature	57
Table 5.7: System Test Cases	60
Table 5.8: Confusion Matrix	61
Table 5.9: Summary of Class-wise metrics	62
Table 5.10: System Performance metrics	65

CHAPTER 1

INTRODUCTION

1.1 Introduction

The Futsal bookings and management platform is an innovative web-based application designed to facilitate real-time futsal court booking and management, with a primary focus on digitizing the operations of futsal centers across Nepal. It allows users to discover and connect with futsal facilities across different locations, creating opportunities for seamless court reservations, live slot availability checks, and community engagement through ratings and reviews. The platform is designed to be intuitive and user-friendly, enabling players to easily search for courts, book sessions, and manage their reservations through a centralized digital interface.

For players, this platform offers features such as AI-powered Sentiment Analysis of reviews , personalized court recommendations based on location, and booking history tracking, allowing users to monitor their activity and make informed decisions. The platform also supports social interaction through community ratings, reviews, and a discussion forum, fostering a collaborative and trustworthy sports community.

From the administrative perspective, Futsal Bookings and management system provides a robust backend system for futsal owners to manage courts, time slots, pricing, and customer interactions. Administrators can oversee platform-wide activity, monitor tenant usage, and ensure smooth operation of the system. The system leverages modern technologies such as the MERN stack, Redis, and Docker to provide reliable, scalable, and secure booking capabilities within a multi-tenant SaaS architecture.

One of the key advantages of Futsal bookings and management platform is its emphasis on intelligent features and data-driven decision support, allowing futsal owners to access analytics dashboards with revenue trends, peak usage times, and customer engagement insights.

Moreover, Futsal Bookings and Management is developed with future scalability in mind, enabling integration with additional features such as mobile application support, online

payment gateways, tournament management, and expanded AI capabilities for enhanced user experience.

1.2 Problem Statement

The growing demand for futsal facilities in urban areas of Nepal has highlighted several challenges for both players and facility operators. Traditional booking methods, such as phone calls, handwritten registers, and social media messages, often fail to provide real-time availability updates and lead to frequent double bookings and scheduling conflicts.

Players frequently struggle to find available courts at convenient times, and opportunities to compare multiple futsal centers based on location, pricing, and user reviews are extremely limited. The absence of a centralized digital platform makes the process of discovering and booking a suitable futsal court time-consuming and inefficient.

From the facility management perspective, futsal owners face challenges in tracking bookings, managing schedules, and maintaining accurate records of revenue and customer interactions. The lack of digital tools also limits their ability to make data-driven decisions, as insights such as peak usage times and revenue trends are not readily available.

1.3 Objectives

- To develop a centralized online platform for futsal discovery and real-time court booking, ensuring convenience and accessibility for users.
- To implement a multi-tenant SaaS architecture that allows multiple futsal facilities to operate independently within a single scalable and secure system.
- To enable community features including verified ratings, reviews, and a discussion forum to enhance user interaction and trust.

1.4 Scope and Limitation

The scope of the Futsal Booking and Management System are as follows:

- Enabling users to search futsal centers by location, availability, and pricing.
- Providing real-time court booking and slot conflict prevention.
- Managing reservation history, cancellations, and booking confirmations.
- Allowing futsal owners to manage courts, time slots, pricing, and customer interactions through a dedicated dashboard.
- Providing an admin panel for platform-wide operations, tenant management, and subscription plans.
- Incorporating community features such as ratings, reviews, and discussion forums.
- Integrating a location-based recommendation engine to suggest nearby futsal centers based on user proximity and preferences.

Every system has errors so this system also has limitations which are as follows:

- Dependence on internet connectivity, the system may not function properly without a stable internet connection.
- Mobile application support is limited in the initial phase, as the platform is primarily designed for desktop and web browsers.
- AI-powered features are based on available training datasets and may not capture every nuance of user sentiment accurately.
- The system's effectiveness depends on futsal owners actively maintaining up-to-date court information and availability schedules.
- Requires user training for effective usage of owner dashboards and admin features.
- Currently limited to futsal facility management; support for other sports may be added in future updates.

1.5 Development Methodology

The Futsal bookings and management platform is developed using the **Agile Development Methodology**, emphasizing flexibility, collaboration, and iterative progress. The development process is organized into sprints, where each sprint focuses on delivering key functionalities while ensuring a user-centric design.

Iteration 1: The first phase establishes core infrastructure, including environment setup, database schema design, and wireframe creation. A minimum viable product (MVP) is developed covering user registration, login, and basic futsal listing.

Iteration 2: This iteration focuses on the core booking engine, including slot availability checks, conflict prevention algorithms, booking confirmation, and cancellation management.

Iteration 3: The third iteration integrates the futsal owner dashboard with court management, pricing configuration, schedule management, and booking approval workflows.

Iteration 4: In this phase, community features including ratings, reviews, and discussion forums are implemented alongside notification systems and user profile management.

Iteration 5: The final iteration introduces advanced AI and ML components including sentiment analysis and location-based recommendations. The system is fully tested for performance, security, and scalability.

By following this iterative Agile methodology, futsal management and bookings platform ensures continuous improvement, rapid delivery of features, and adaptability to feedback throughout the development process.

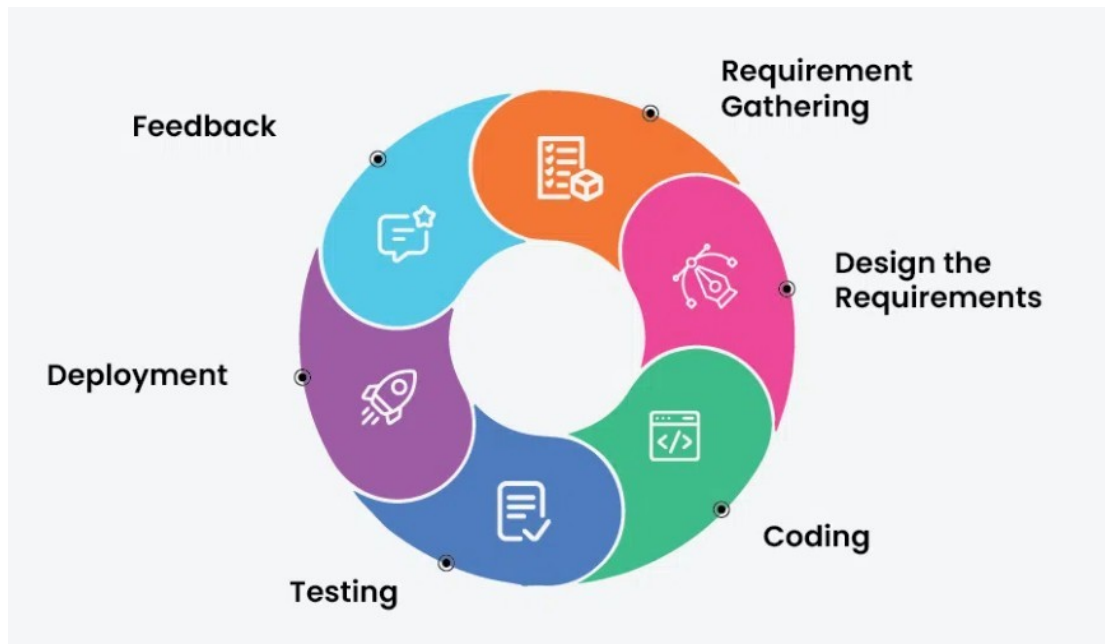


Figure 1.1 Agile Development model

1.6 Report Organization

The Report Organization describes how the study is structured and conducted. This report is divided into six chapters, each addressing different aspects of the study. Chapter One presents the introduction, statement of the problem, objectives of the study, scope and limitations, development methodology and organization of the study. Chapter Two focuses on the background study and review of related literature, providing the necessary theoretical foundation. Chapter Three discusses system analysis, including requirement analysis, feasibility study, data modeling, and process modeling of the system. Chapter Four deals with the design of the system, particularly the system architecture and its components. Chapter Five explains the implementation phase, the algorithms used, and the tools applied, along with the testing of the system both as individual units and as a 3complete system. Finally, Chapter Six provides the conclusions drawn from the study, highlights its limitations and offers recommendations for future improvements.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The need for an Online Futsal Booking and Management System arises from the limitations of traditional manual booking methods such as phone calls, handwritten registers, and social media messages. These older approaches often cause double bookings, scheduling conflicts, missing records, and difficulties in tracking court availability and customer interactions. As the number of futsal centers in Nepal continues to grow rapidly, managing bookings manually becomes increasingly inefficient and unreliable.

A digital futsal booking platform addresses these issues by automating the entire reservation process. Players can search for available courts online, make bookings in real time, and receive instant confirmation, while the system prevents slot conflicts through automated time-slot validation. Futsal owners benefit from dedicated dashboards that streamline court management, pricing configuration, and customer communication, reducing administrative workload and minimizing human error.

Additionally, the platform integrates machine learning components to enhance user experience through sentiment analysis of reviews and location-based court recommendations. Overall, the Futsal system enhances booking efficiency, data accuracy, and operational transparency for both players and futsal facility operators.

2.2 Literature Review

The Futsal Booking and Management System is a web-based SaaS application designed to automate the process of discovering, booking, and managing futsal courts. It enables players to search for futsal centers by location and availability, make real-time reservations, and manage their booking history, while futsal owners can manage courts, schedules, pricing, and customer interactions through a dedicated management dashboard. The system maintains a centralized

multi-tenant database that stores each facility's booking records, customer data, and operational details, ensuring that all information remains accurate, up-to-date, and securely isolated per tenant [1].

Traditionally, futsal centers managed bookings through phone calls, paper registers, and messaging apps, which were inefficient, time-consuming, and prone to double bookings, data loss, and poor record management [2]. With advancements in web technologies and digital platforms, automated booking systems have replaced manual processes, improving reliability, scheduling clarity, and operational transparency across the sports facility industry [3]. A study by Suhaimi and Sahid developed a futsal court booking system to replace manual records with a database-driven platform, demonstrating measurable improvements in booking visibility and electronic data management through a structured booking interface built using the prototyping model [1]. Similarly, Fauzi et al. presented a web-based futsal field rental information system aimed at improving business opportunities and operational efficiency in the sports rental sector, highlighting that website-based booking platforms help replace handwritten rental processes while increasing accessibility and customer reach [2].

Industry analysis further supports the effectiveness of specialized sports booking software, reporting measurable reductions in scheduling conflicts and improved operational transparency when centralized booking systems are deployed [3]. In addition, mobile-first and Progressive Web Application (PWA) approaches are recommended for booking platforms because they provide a responsive, app-like user experience across devices without the high cost of native mobile app development [4]. Modern system design literature recommends multi-tenant SaaS architecture for shared booking platforms serving multiple independent facility owners with proper data isolation, ensuring scalability and security across all tenants [5].

Machine learning techniques are increasingly applied to process user reviews and support decision-making in digital platforms. Term Frequency-Inverse Document Frequency (TF-IDF) is a widely used text feature extraction technique that identifies the most important terms in a document relative to a collection of documents, enabling automatic review summarization and smarter content ranking [6]. Recent studies utilize Logistic Regression as a robust supervised learning algorithm for sentiment analysis across various domains, classifying user feedback into binary outcomes such as positive or negative sentiment based on extracted TF-IDF feature vectors [7]. Furthermore, location-based recommendation methods using coordinate-based

distance computation such as the Haversine formula are used to rank nearby services based on user proximity and preference patterns, improving the relevance of court suggestions presented to users [6][7].

The Futsal Bookings and management system is developed following the Agile Software Development Life Cycle (SDLC), which supports iterative development, continuous feedback, and flexible requirement handling throughout the project lifecycle [8]. Each sprint delivers a working version of the system with improved features and refinements, ensuring that modules such as booking management, owner dashboards, and machine learning components are developed and tested incrementally [8].

In conclusion, the AllFutsal system serves as a secure, scalable, and intelligent platform that digitizes traditional futsal booking processes and brings efficiency to sports facility operations. It not only enhances scheduling accuracy, transparency, and operational management for futsal owners but also provides AI-driven insights and personalized recommendations for players, addressing the clear gap in existing manual and partially digital solutions.

CHAPTER 3

SYSTEM ANALYSIS

3.1 System Analysis

System analysis involves studying the system in detail to understand how it works and what improvements are needed. It focuses on identifying problems, gathering data, and separating the system into smaller parts to make it easier to study. By analyzing each part and how they connect, we can clearly see what the system is expected to do. In this way, system analysis acts as a method of problem solving, helping to check whether each component works properly and contributes toward the overall goal of the system.

3.1.1 Requirement Analysis

To develop the futsal bookings and management, we identified both functional and non-functional requirements that the system must fulfill.

3.1.1.1 Functional Requirement

Functional requirement are the features and functions that the Futsal Bookings and management System must provide.

- **User Registration and Authentication:** Only registered users can access the system. Users can create and manage their profiles, while futsal owners and administrators have extended privileges to manage courts, bookings, and platform-wide operations respectively.
- **Futsal Discovery and Search:** Users can search for futsal centers by location, price, and available time slots, and view detailed information including court facilities, pricing, and community reviews before making a booking decision.

- **Real-Time Court Booking:** Users can book available futsal courts online, receive instant booking confirmation and notifications, and manage their reservation history. Bookings can also be cancelled or rescheduled based on the platform's defined rules and policies.
- **Ratings, Reviews and Recommendations:** Users can submit ratings and reviews after playing, and the system provides personalized futsal court recommendations based on the user's location and preferences using a location-based recommendation engine.
- **Futsal Owner Management:** Futsal owners can register and manage their futsal center profile, add and manage courts, set available time slots and pricing, approve or manage incoming booking requests, view booking schedules and customer details, manage offers and subscription plans, view an analytics dashboard for bookings and revenue insights, and respond to customer reviews and queries.
- **Admin Control Panel:** Administrators can manage all futsal tenants on the platform, manage user accounts and permissions, monitor overall system usage and generate reports, manage subscription plans and packages, and handle complaints and system-level moderation to ensure smooth platform operation.
- **Community Forum:** Registered users and futsal owners can create, view, and participate in forum discussions related to futsal topics such as match arrangements, court reviews, team formations, and general sports discussions, fostering a collaborative community environment.
- **Notification and Confirmation System:** The system automatically sends booking confirmations, status updates, and relevant alerts to users, futsal owners, and administrators based on actions performed within the platform.

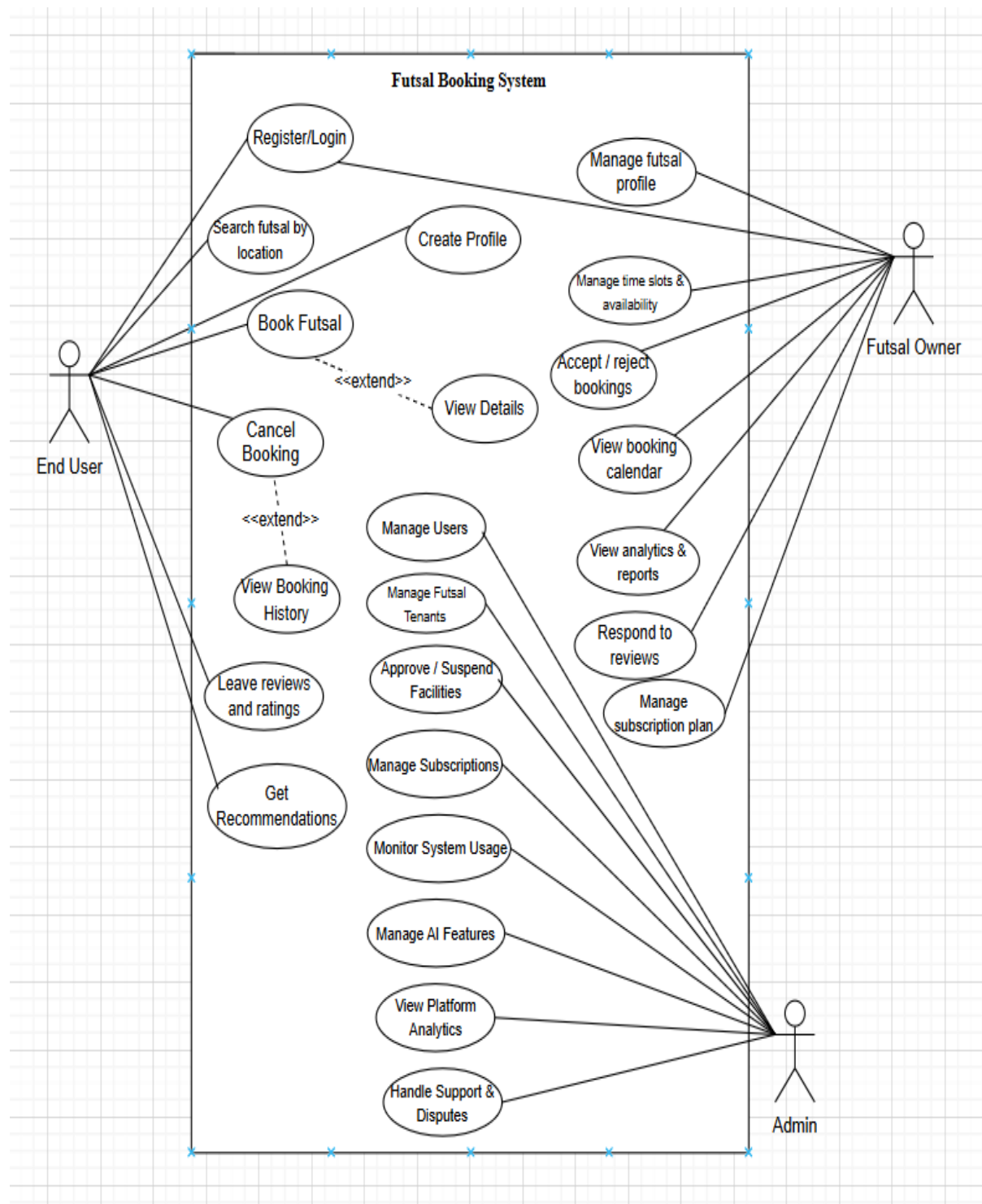


Figure 3.1 : Use Case Diagram of Futsal Bookings Management System

3.1.1.2 Non-Functional Requirements

Non-functional requirements are the requirements that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors. They are contrasted with functional requirements that define specific behavior or functions. The non- functional requirements of the system are:

- **Usability:** The interface should be simple, clean and easy to use for all users (admin, teacher and student).
- **Reliability:** System is accurate, as we don't use many components and thus provides the user specified results in specific time duration.
- **Performance:** System should handle multiple user requests simultaneously without slowing down.
- **Scalability:** The system should be designed so it can accommodate more users or institutions in the future without major redesign.
- **Security:** All sensitive data must be securely encrypted during storage and transmission. Only authorized users can access certain features, and all communication must be encrypted using HTTPS.
- **Maintainability:** The system should be easy to update and maintain, with a clear and modular architecture that supports future enhancements and feature upgrades.

3.1.1.3 Dataset Description

For this project, a custom synthetic dataset was created to test and validate the Futsal Booking and Management System workflow, particularly for sentiment analysis of user reviews and TF-IDF based review summarization. The dataset contains 1200 records with 5 attributes, designed to simulate real-world futsal court user review submissions. The dataset was synthetically generated using a Python script due to the unavailability of domain-specific public futsal review datasets on platforms such as Kaggle and Hugging Face. In addition, the review scenarios and rating patterns were prepared based on discussions among the three project team members, who are regular futsal players, and were further validated by the project supervisor to ensure realistic and practical review data representation.

id	user_id	rating	review	sentiment_label
1	461	3	Well Spacious court and proper lighting setup at night session	Positive
2	301	2	To be honest Facilities were not clean at all but during our regular game	Negative
3	197	3	Frankly speaking Spacious court and proper lighting setup at night session	Positive
4	152	5	Overall Very professional staff and management during our match time	Positive
5	443	3	To be honest Facilities were outdated and not maintained we played in the evening	Negative
6	427	4	Actually The court was spacious and well-maintained, providing ample room for playe	Positive
7	318	1	To be honest Not satisfied with the overall experience during our match time	Negative
8	404	4	The best place to play futsal in town but staff was rude and unhelpful while playing w	Positive
9	383	4	From what I saw The turf was in excellent condition and very safe but poor lighting an	Positive
10	272	4	From my experience The booking process was seamless and efficient, making it easy	Positive
11	496	2	To be honest Too much delay in booking confirmation while playing with friends	Negative
12	387	3	Actually The booking process was seamless and efficient, making it easy to reserve a	Positive
13	188	5	Well during peak hours	Positive
14	136	1	horrible	Negative
15	482	3	Based on my visit Too much delay in booking confirmation on a rainy day	Negative
16	168	3	Well The court was overcrowded and chaotic but the price was reasonable for the qu	Negative
17	155	5	From my experience The best place to play futsal in town during weekend booking	Positive
18	253	4	Overall Loved the turf and lighting setup while playing with friends	Positive
19	70	3	not satisfied	Negative
20	80	5	In my opinion Affordable price with great quality during our regular game	Positive
21	17	2	Well Had to wait for a long time to get the court ready during weekend booking	Negative
22	51	4	From my experience Spacious court and proper lighting setup while trying out the new	Positive
23	496	1	In my opinion Not satisfied with the overall experience but booking system was fast a	Negative
24	340	3	never again	Positive
25	65	3	Well Great futsal court with excellent facilities but the worst futsal court i've ever been	Positive
26	169	1	From my experience Not worth the price during weekend booking	Negative
27	67	1	Court was dirty and poorly maintained while testing the lighting	Negative

Figure 3.2 : Dataset records

The attributes include both categorical and numerical features describing each review record. The attributes are as follows:

- **id:** Unique identifier for each user review record.
- **user_id:** Unique identifier for the user who submitted the review.
- **rating:** Numerical rating provided by the user on a scale of 1 to 5, where 1–2 indicates a negative experience and 4–5 indicates a positive experience.

- **review:** The textual content of the user's review submitted after playing, used as input for TF-IDF feature extraction and sentiment classification.
- **sentiment_score:** A numerical probability score between 0.00 and 1.00 representing the confidence of the sentiment prediction, where values closer to 1.00 indicate stronger positive sentiment.

For sentiment classification, the sentiment category was derived from the rating values during preprocessing, where lower ratings represented negative sentiment and higher ratings represented positive sentiment. The review text data was preprocessed by removing stop words and applying tokenization before TF-IDF vectorization to make the textual data suitable for machine learning analysis.

This dataset was synthetically prepared to simulate realistic futsal user review scenarios, providing a practical basis for testing the TF-IDF Review Feature Extraction Algorithm and the Logistic Regression Sentiment Classification Algorithm implemented within the Futsal Bookings and Management platform.

3.1.2 Feasibility Analysis

The proposed project's viability is determined by the feasibility study:

I) Technical Feasibility

The futsal bookings and management platform is a web-based application that uses a multi-tenant SaaS architecture to manage futsal court bookings, owner dashboards, and AI-powered review analysis across multiple independent futsal facilities. The system incorporates a Slot Conflict Prevention Algorithm for real-time booking validation, a Haversine-based Location Distance Algorithm for court recommendations, and machine learning components including TF-IDF feature extraction and Logistic Regression for sentiment classification of user reviews. Since the system is built using widely adopted and well-documented technologies such as React, Node.js, Express.js, and MySQL, and does not require any specialized external hardware for functioning, we will be able to construct this system using resources freely available on the internet and our existing knowledge of modern web development technologies. Machine learning components can be implemented using Python libraries such as Scikit-learn, with training datasets prepared synthetically and processed locally without the need for expensive computing infrastructure.

ii) Operational Feasibility

The futsal bookings and management system aims at making it easier for players to discover and book futsal courts in real time, and for futsal owners to manage their courts, schedules, pricing, and customer interactions through a dedicated dashboard. For the system to be effective, it must be accessible to players, futsal owners, and administrators through any standard web browser on a desktop or laptop device without requiring any additional software installation. After deployment, the system can be used with minimal training since the interface is designed to be intuitive and user-friendly, closely resembling familiar online booking platforms that users already interact with in their daily lives. Futsal owners are only required to register their facility and maintain up-to-date court availability information for the system to operate effectively. Apart from this, no prerequisite technical knowledge is required from end users to operate the platform.

iii) Economic Feasibility

The futsal system will be constructed at a minimum cost. Almost every resource required for the development of the system, including frameworks, libraries, development tools, and machine learning packages, will be acquired freely from the internet as open-source software. The MERN stack technologies used for development are completely free and open source, and the machine learning components will be implemented using freely available Python libraries such as Scikit-learn and NLTK. For this academic project, free-tier cloud hosting services and local development environments will be used, making the overall development cost negligible. We will be able to complete this project using our existing understanding of available programming languages, frameworks, and technologies without requiring any significant financial investment.

iv) Schedule Feasibility

Phase	Task	Duration (in weeks) 2082													
		1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th
1	Requirement Analysis	■	■												
2	System Design		■	■											
3	Docker Environment Setup			■											
4	Database and Redis Setup				■										
5	Backend Development					■	■	■	■						
6	Frontend Development							■	■	■	■				
7	ML Models Development										■	■			
8	System Integration												■	■	
9	Testing											■	■	■	
10	Documentation				■	■	■	■	■	■	■	■	■	■	■

Figure 3.3 : Gantt Chart

The Futsal Bookings and Management project is planned to be completed within a 14-week timeline in the year 2082, divided into 10 structured development phases. The project begins with Requirement Analysis in weeks 1 and 2, followed by System Design in weeks 2 and 3, and Docker Environment Setup in week 3. Database and Redis Setup is carried out in week 4, after

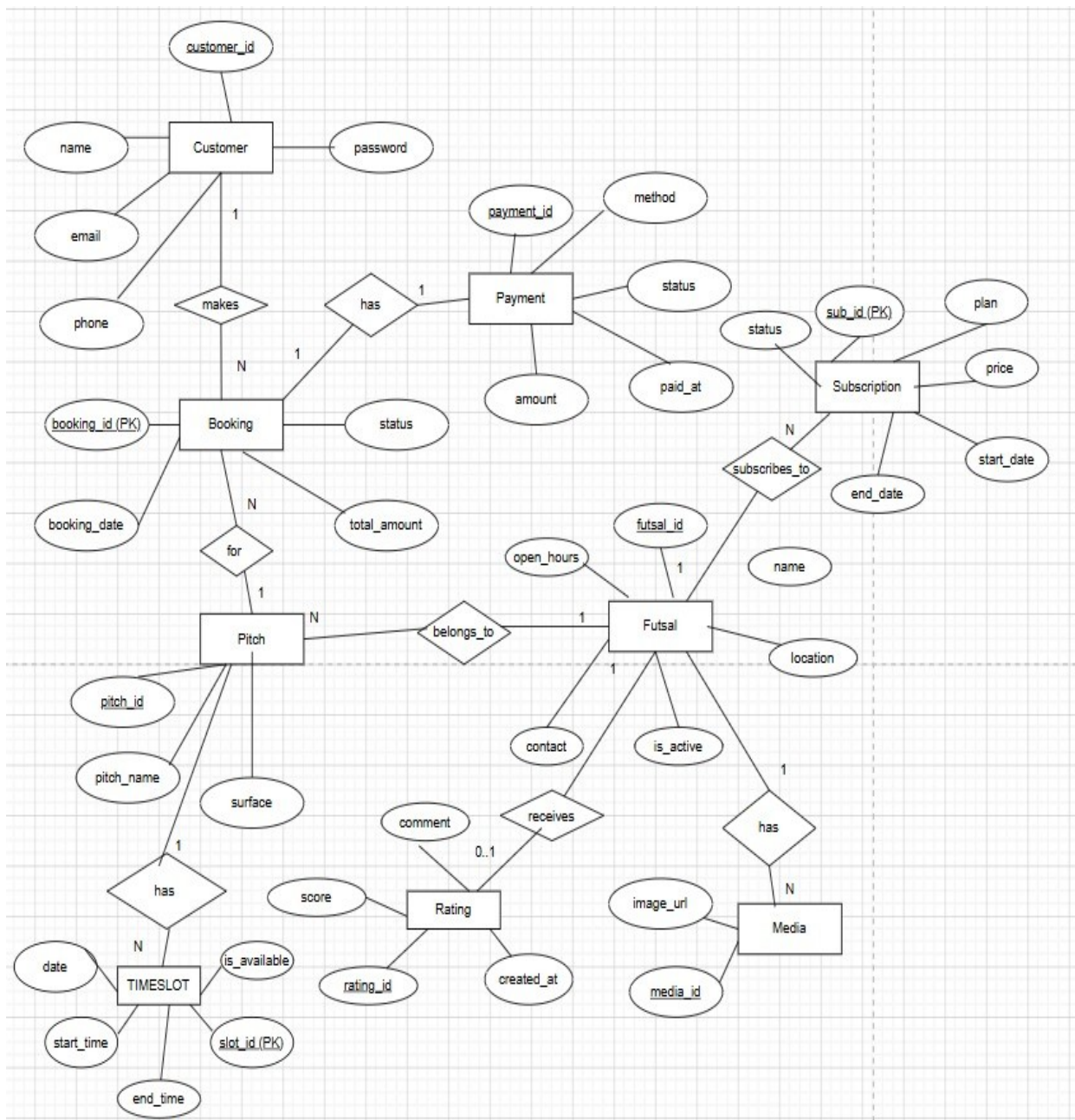


Figure 3.5 : Er Diagram 2

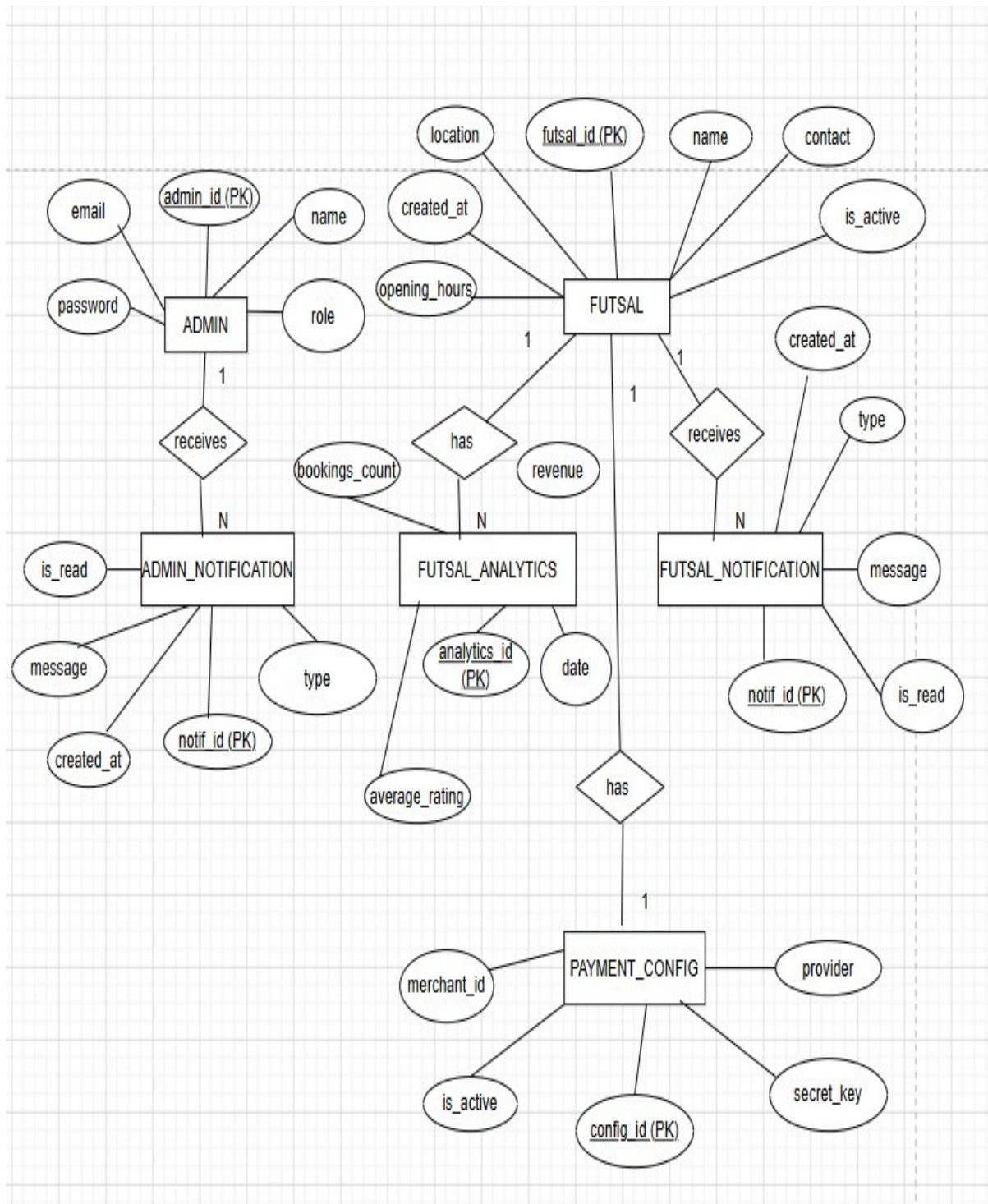


Figure 3.6 : Er Diagram 3

3.1.3.2 DFD level-0 Diagram

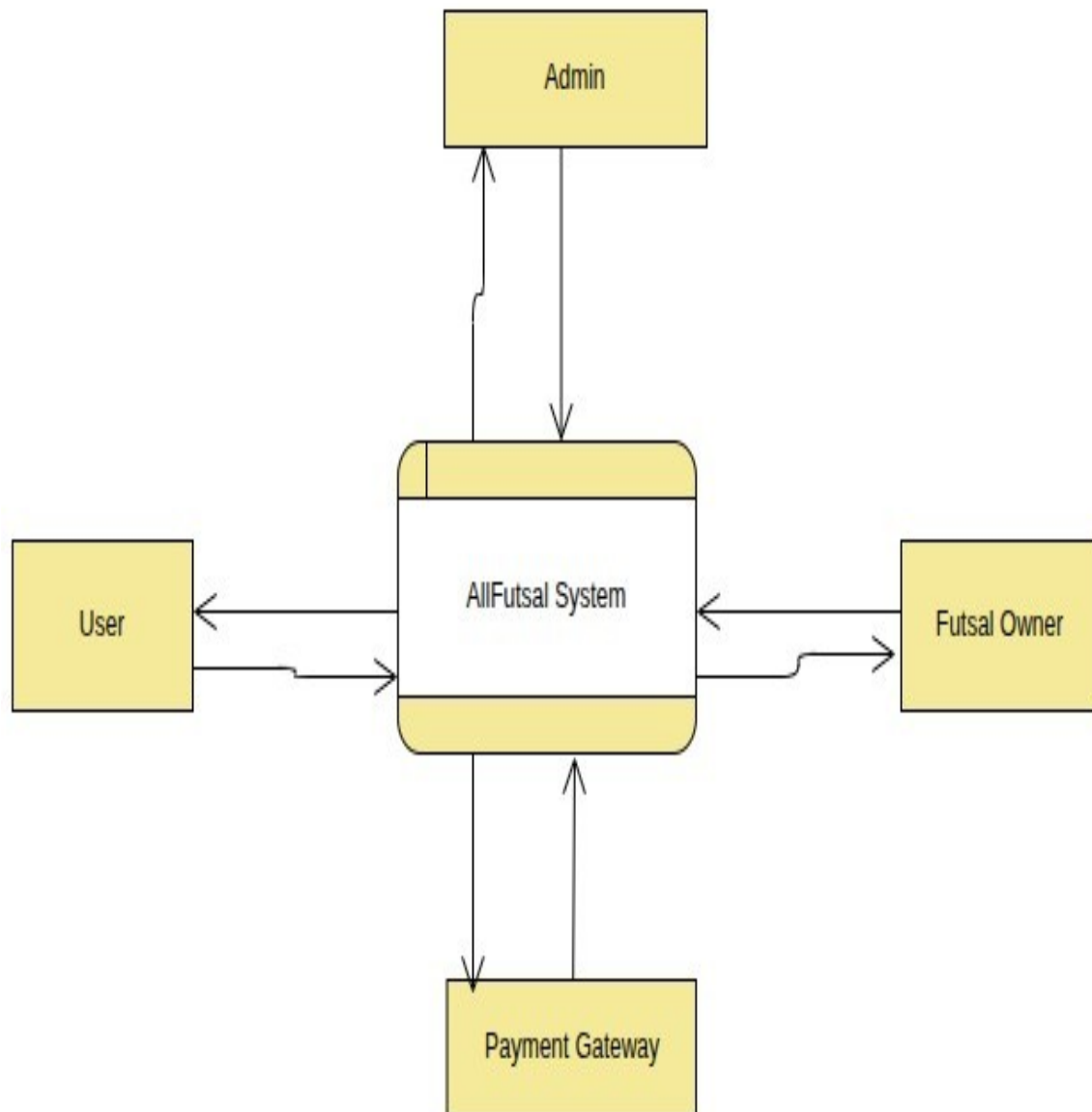


Figure 3.7 : DFD level-0 Diagram

3.1.3.3 DFD level-1 Diagram

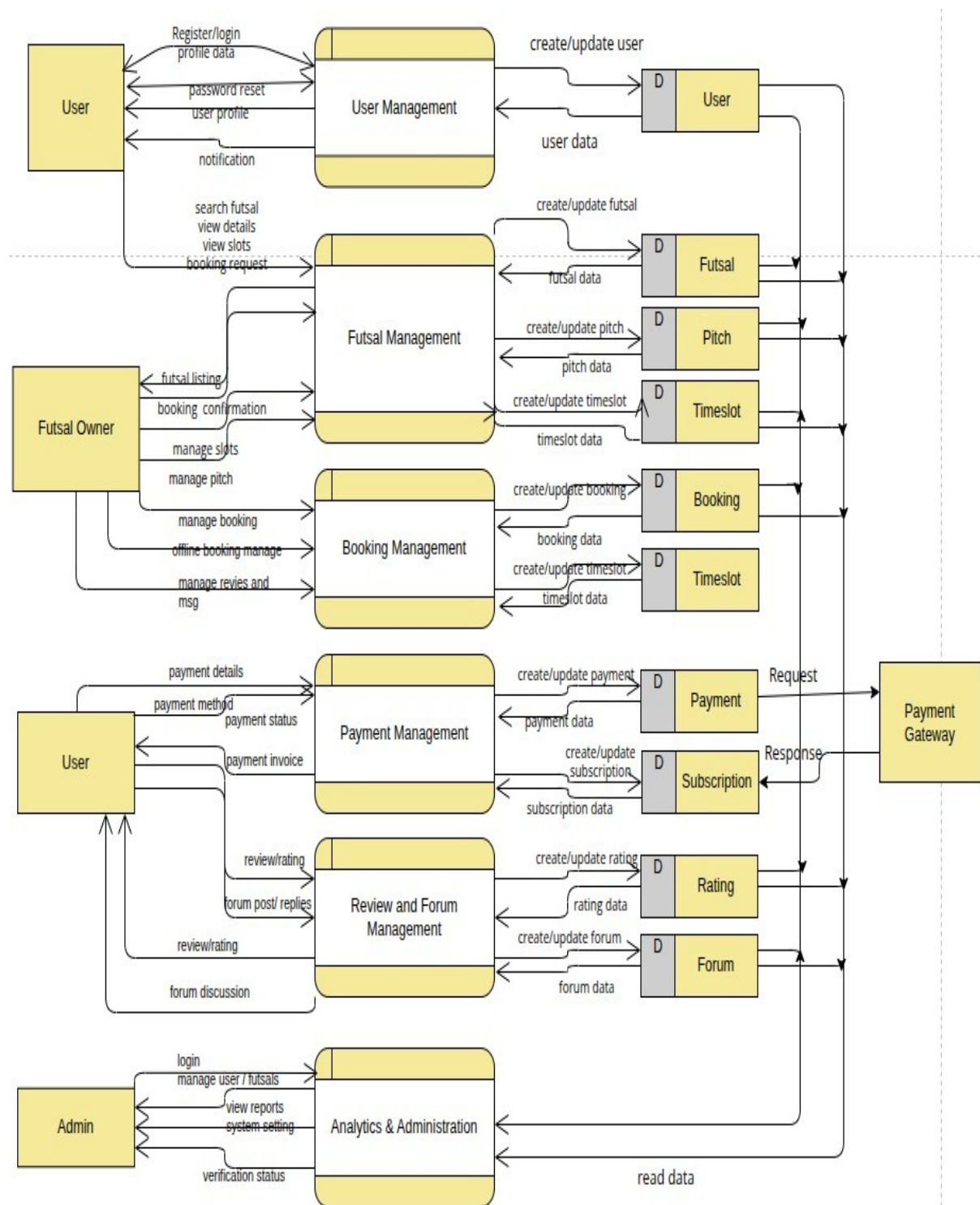


Figure 3.8 : DFD level-1 Diagram

3.1.3.4 DFD level-2 process 1 Diagram

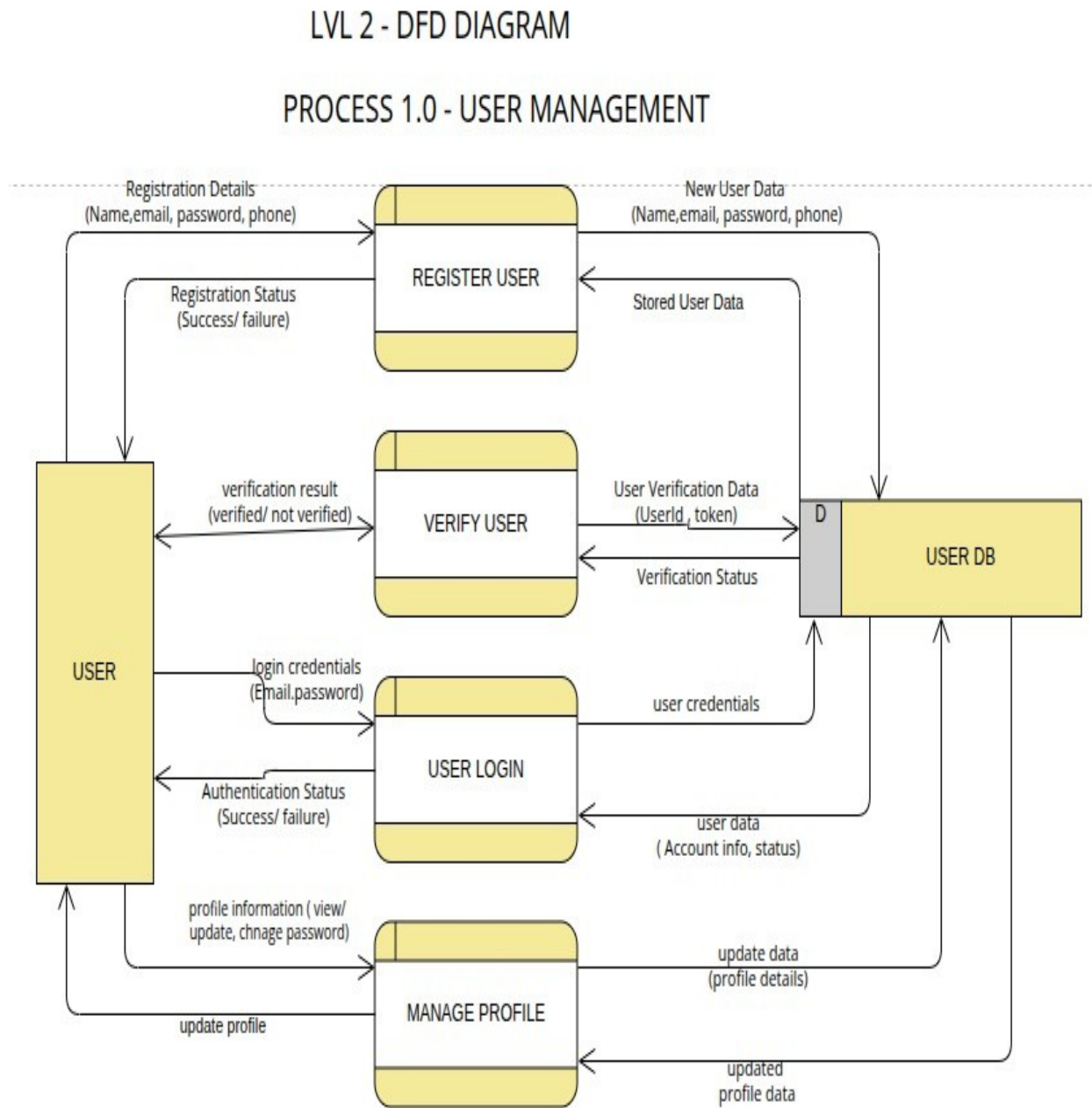


Figure 3.9 : DFD level-2 process 1 Diagram

3.1.3.5 DFD level-2 process 2 Diagram

PROCESS 2.0 - FUTSAL MANAGEMENT

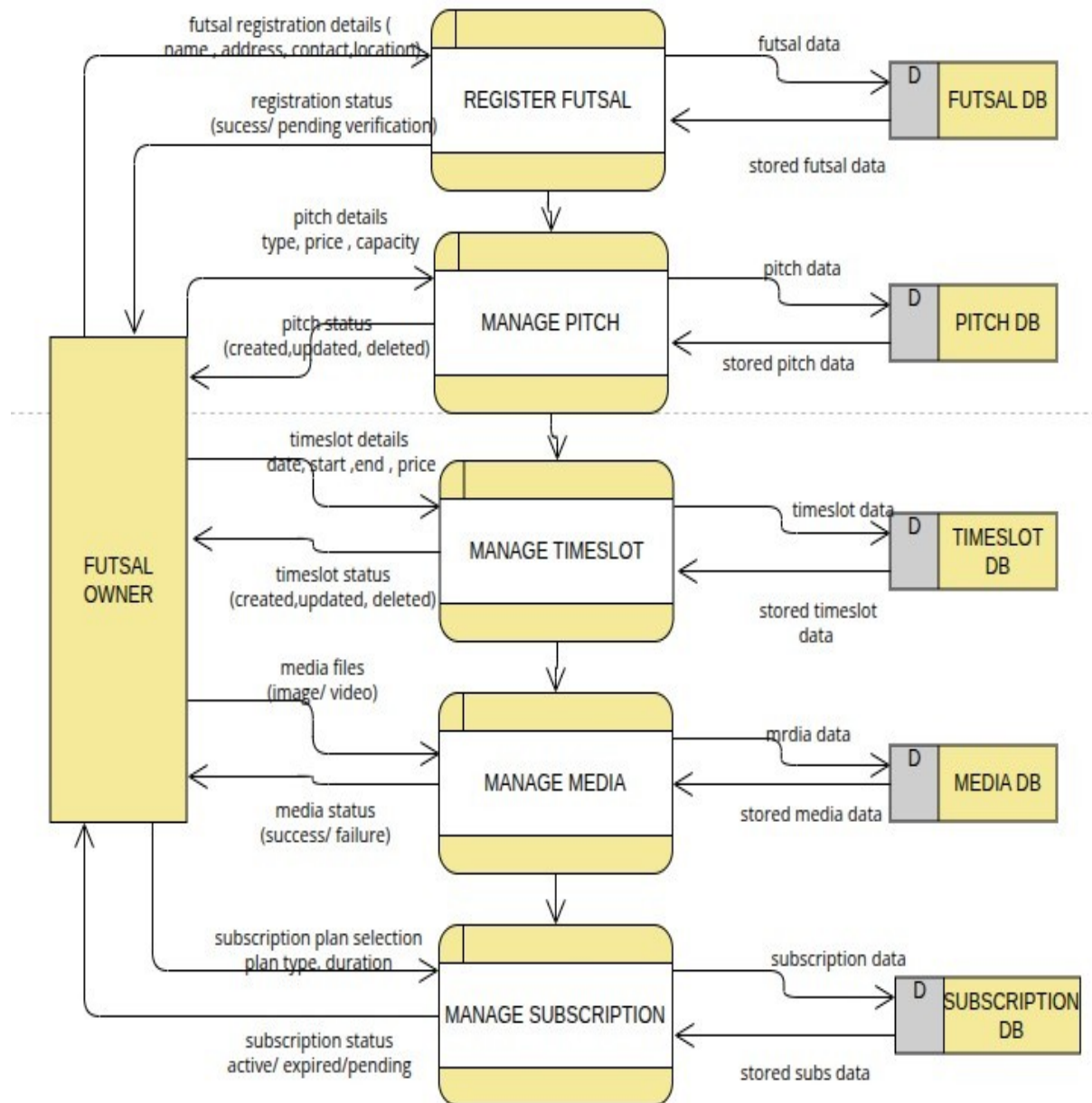


Figure 3.10 : DFD level-2 process 2 Diagram

3.1.3.6 DFD level-2 process 3 Diagram

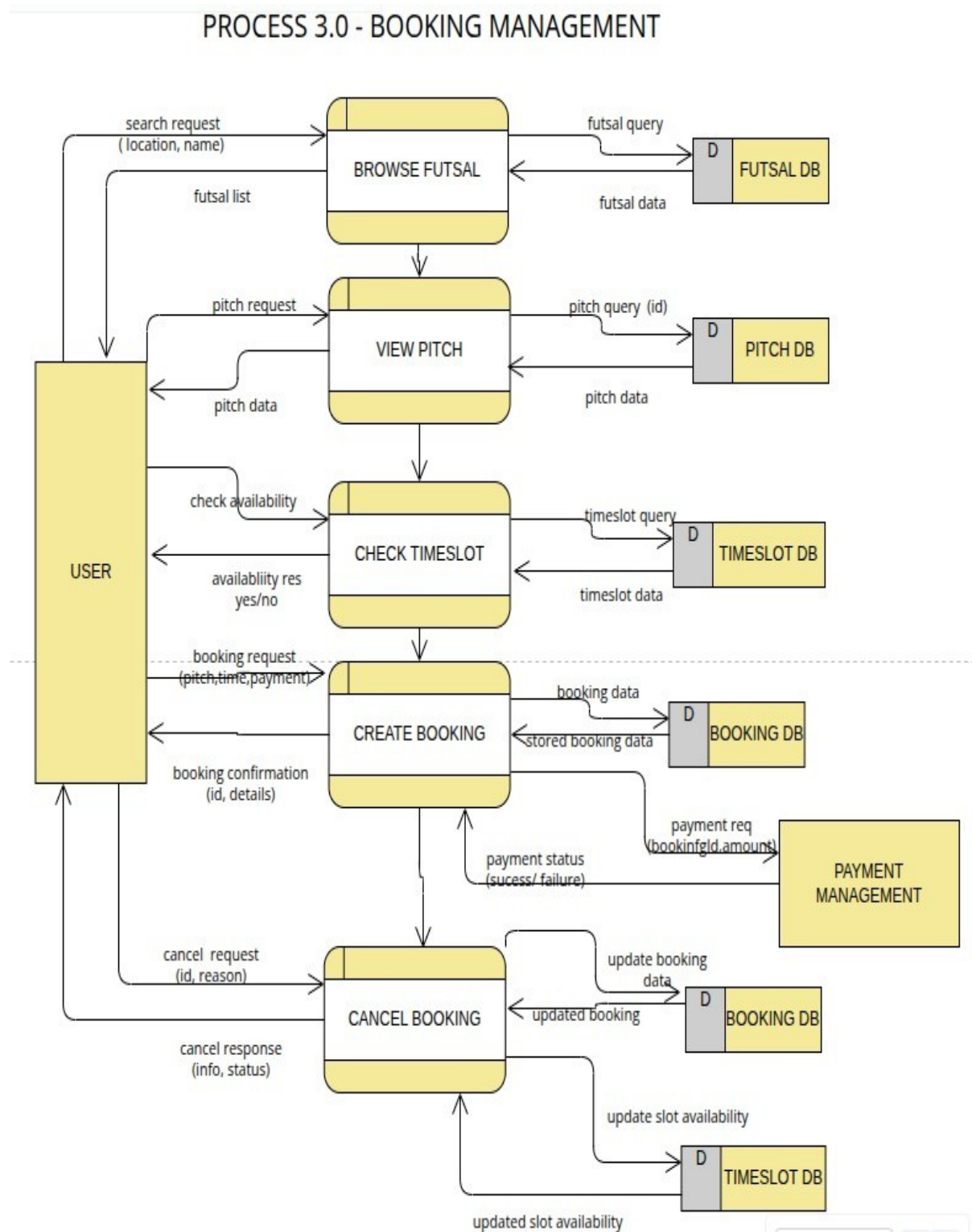


Figure 3.11 : DFD level-2 process 3 Diagram

3.1.3.7 DFD level-2 process 4 Diagram

PROCESS 4.0 - PAYMENT MANAGEMENT

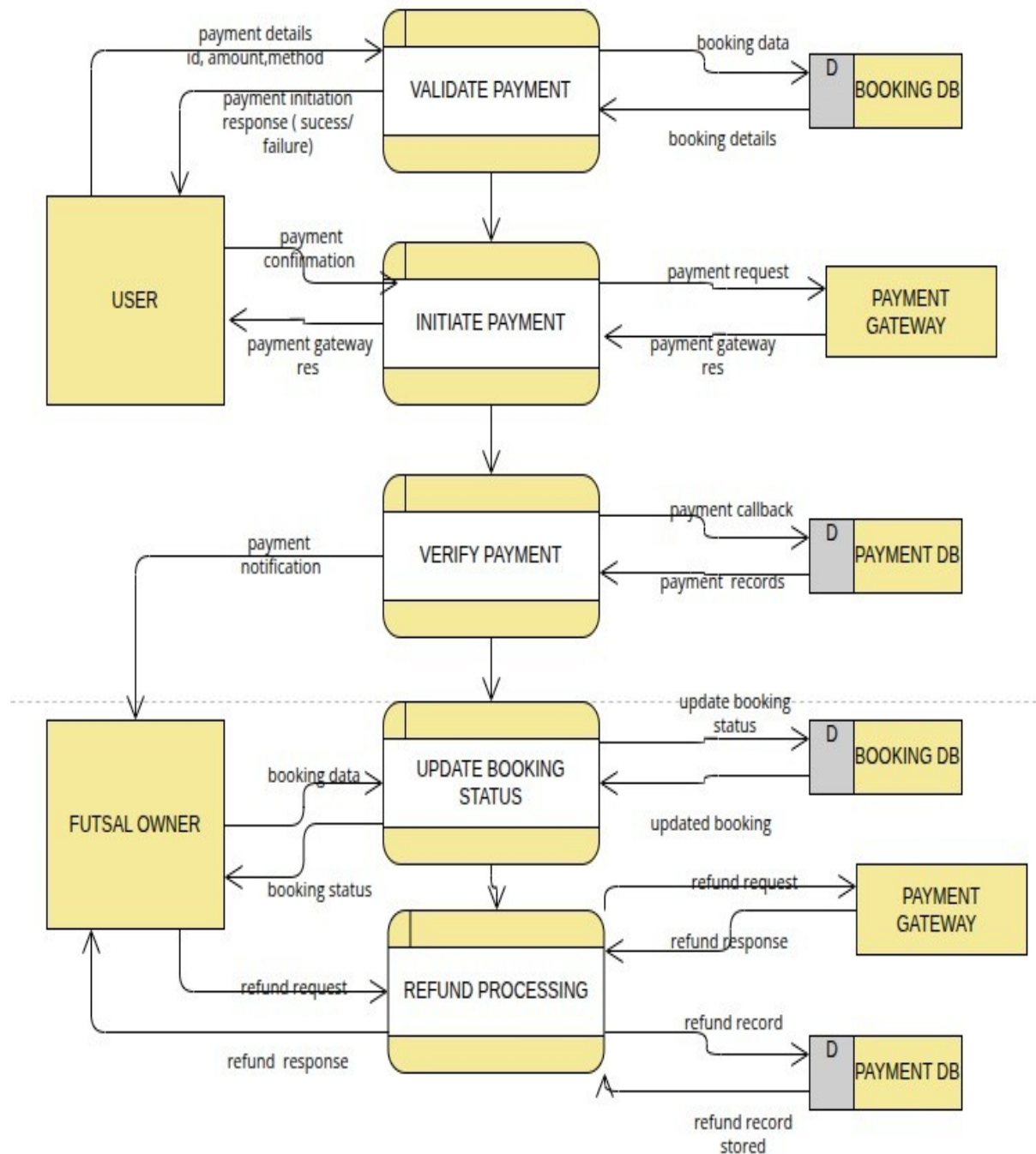


Figure 3.12 : DFD level-2 process 4 Diagram

3.1.3.8 DFD level-2 process 5 Diagram

PROCESS 5.0 - REVIEW & FORUM MANAGEMENT

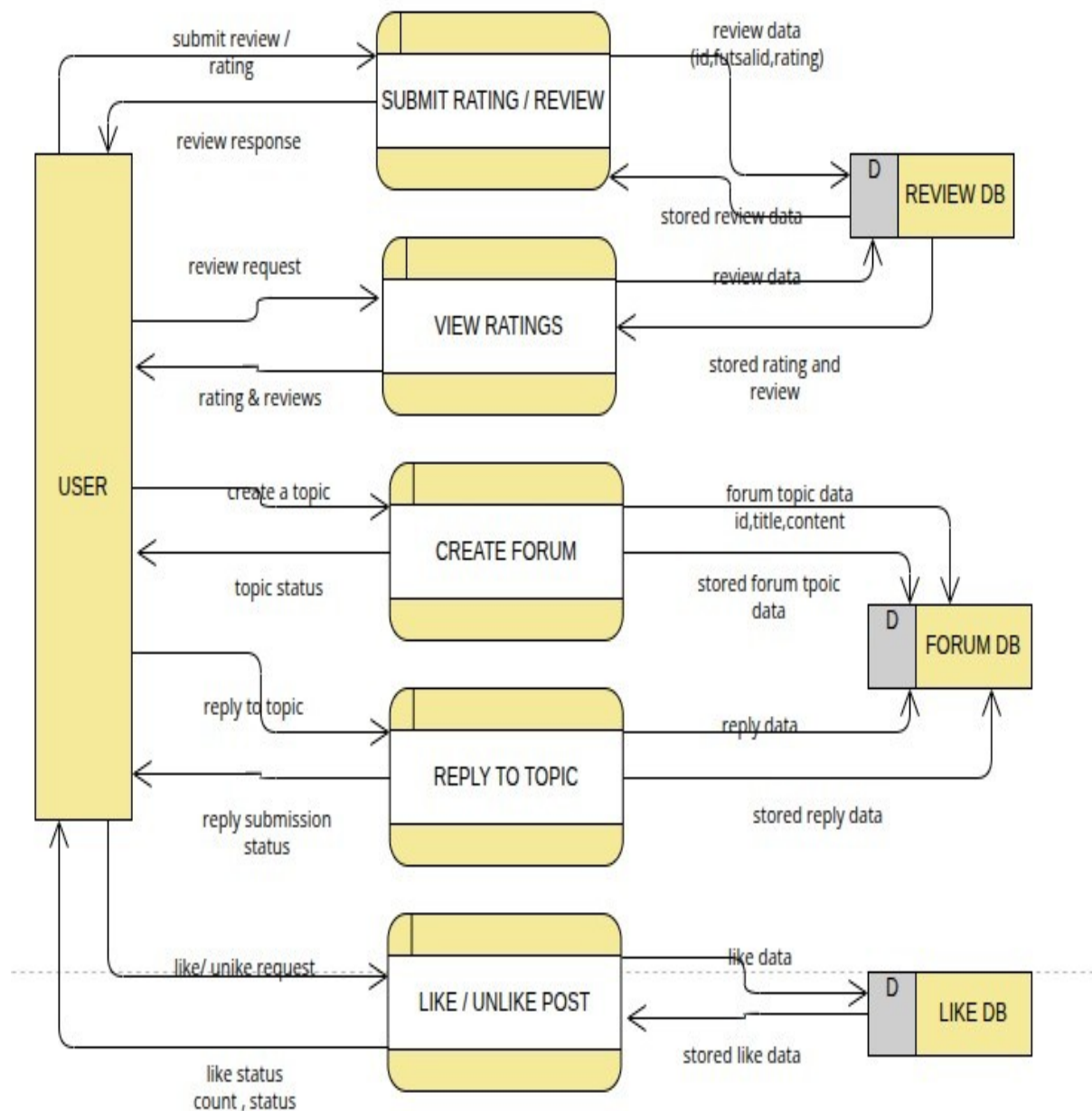


Figure 3.13 : DFD level-2 process 5 Diagram

3.1.3.9 DFD level-2 process 6 Diagram

PROCESS 6.0 - ANALYTICS & ADMINISTRATION PROCESS

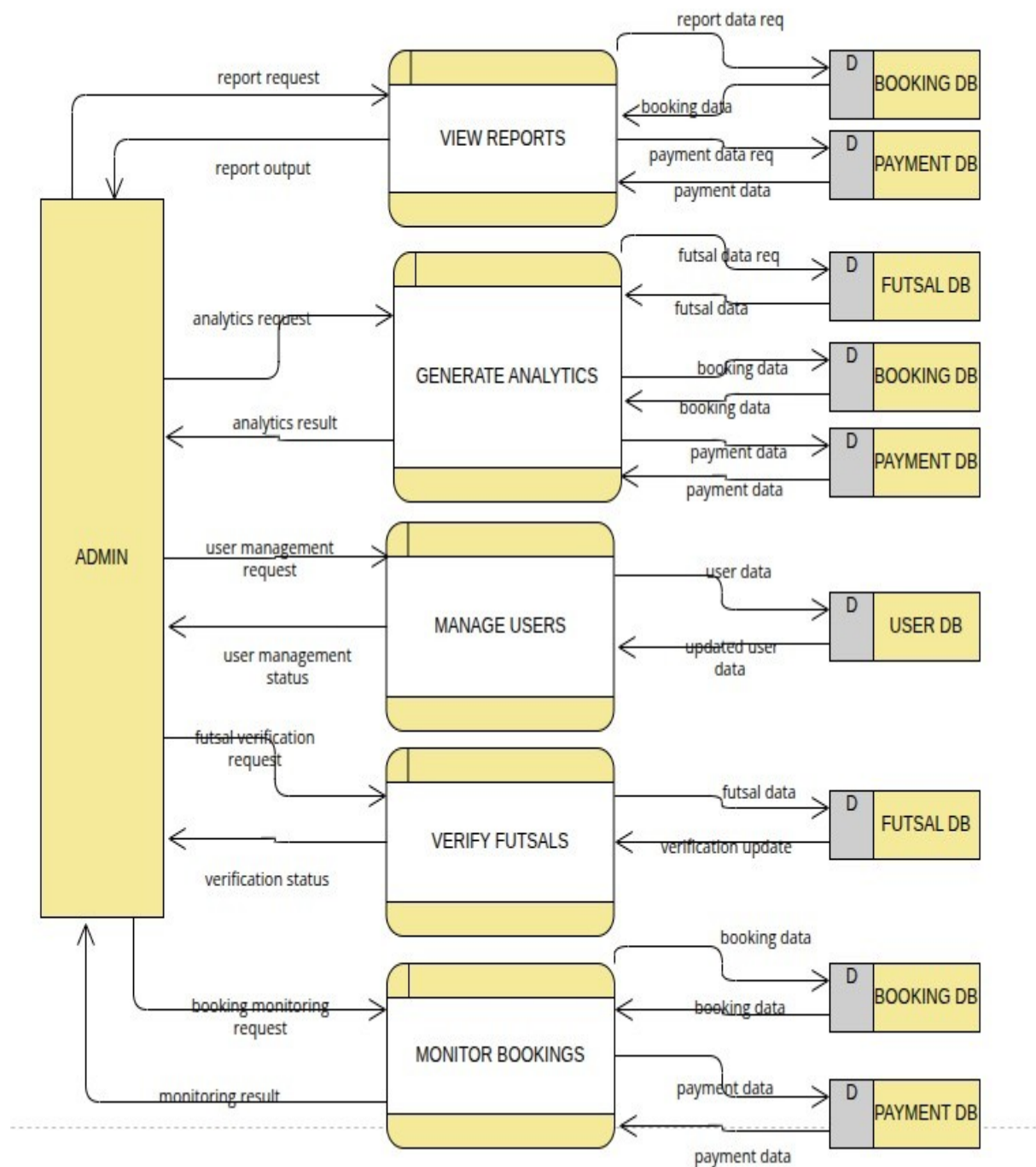


Figure 3.14: DFD level-2 process 6 Diagram

3.1.3.5 Sequence Diagram

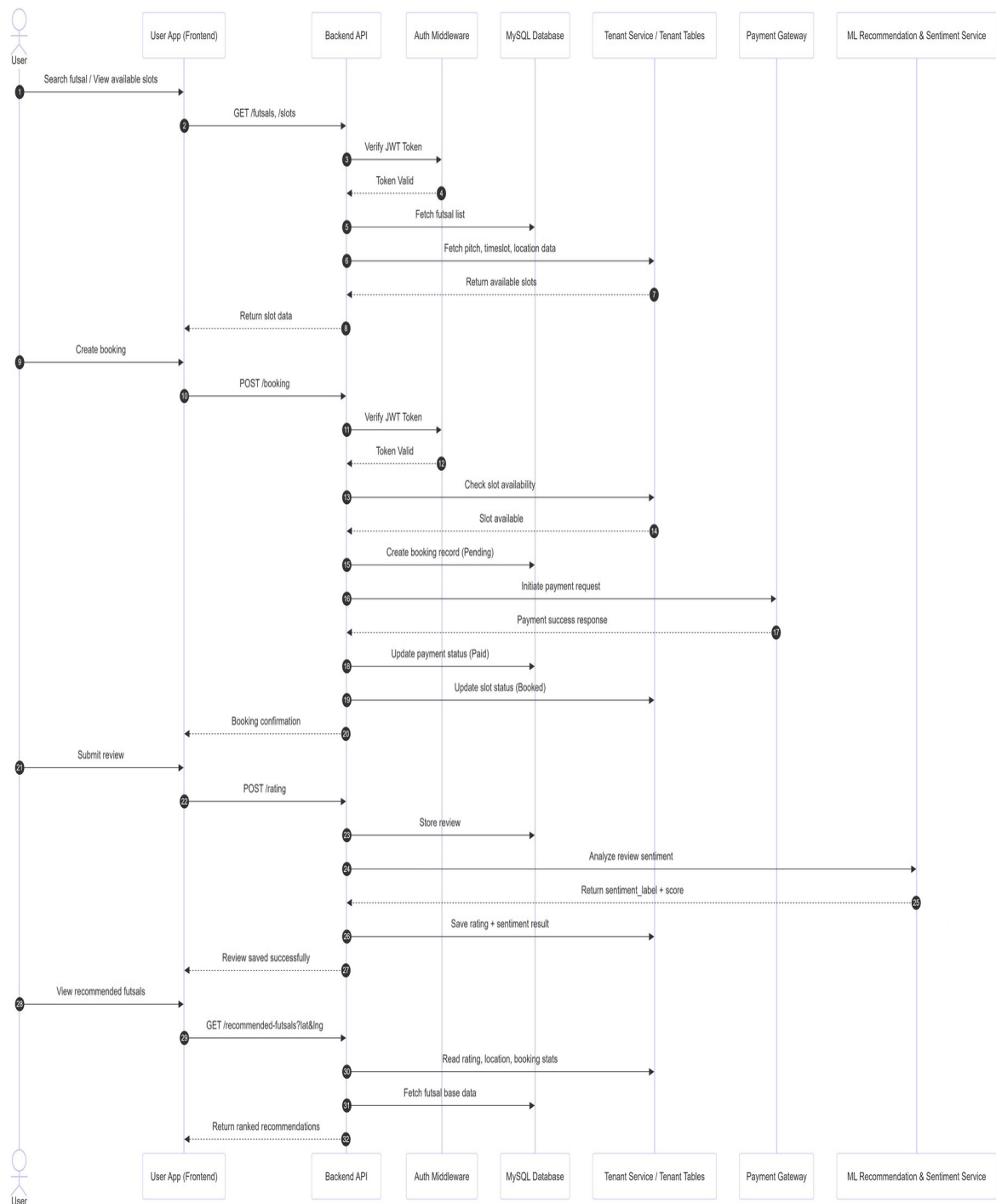


Fig 3.15: Sequence Diagram of Futsal Bookings and Management System

The sequence diagram provides a foundational view of the system's initial design, mapping the critical path from a user's first search to the final generation of a data-driven recommendation. The process is anchored by the Backend API, which serves as the primary bridge between the

user-facing Frontend and the underlying data layers. To ensure a secure starting point for every interaction, the system first routes all requests through the Auth Middleware. Once the user is authenticated, the API performs a synchronized data fetch: it pulls global futsal identity records from the MySQL Database while simultaneously extracting real-time availability and location data from the Tenant Tables. This initial stage establishes a reliable baseline for the user to view accurate slots across various local venues.

The core of the system's operational logic is captured in the final booking and payment phase. When a user submits a booking request, the API conducts a mandatory availability check directly against the Tenant Tables to prevent physical slot conflicts. Following this verification, the system engages the Payment Gateway to process the transaction. Upon a successful response, the architecture ensures data integrity by performing a dual update: the global database logs the financial transaction, while the specific tenant table transitions the slot status to "confirmed." This finalizes the booking loop, providing the user with an immediate confirmation while maintaining an synchronized record of revenue and occupancy across the entire multi-tenant platform.

Beyond the transaction, the diagram outlines the system's ability to transform raw feedback into actionable intelligence. When a User submits a rating, the review is passed to the ML Service, which interprets the text to provide a sentiment label and score. These insights are saved back into the Tenant Tables, directly influencing the venue's standing within the ecosystem. The sequence concludes with the recommendation flow, where the API aggregates these sentiment scores, location data, and booking statistics to rank venues. This final architectural layer demonstrates how the initial inputs of search and booking eventually feed back into the system to drive personalized, intelligent discovery for the community.

CHAPTER FOUR

SYSTEM DESIGN

4.1 Design

System Design includes the means and methodologies to improve the management and control of the software development process. It includes structuring and simplifying the process using several diagrams and standardizing the development process by specifying the required activities and techniques to be implemented.

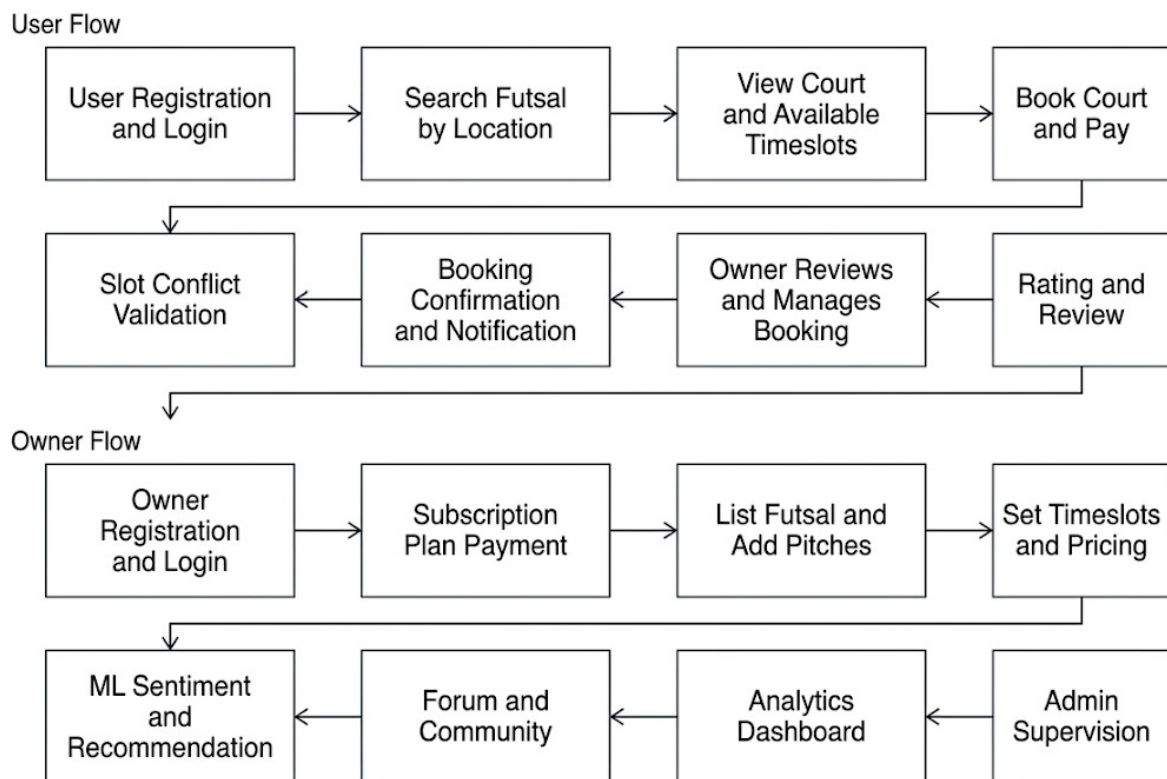


Fig 4.1: High-Level System Design of Futsal Bookings and Management System

The system architecture is shown in Figure 4.1. The procedures involved are:

User Registration and Login

The process begins when a player registers a new account or logs in with existing credentials. During registration, users provide details such as their name, email, and password, with optional Google OAuth social login for convenience. The system validates all inputs, encrypts the password securely, and stores the user record in the database. After a successful login, users access their dashboard to search for futsal centers, manage bookings, and view their booking history.

Search Futsal by Location

Once logged in, users can search for available futsal centers by entering their location or allowing the system to detect their current position. The platform uses the Haversine-based location distance algorithm to calculate proximity between the user and nearby futsal centers, ranking results by distance and relevance. Users can also filter results by price, availability, and ratings to find the most suitable court.

View Court and Available Timeslots

Users can select a futsal center to view detailed information including court facilities, pitch types, pricing, available time slots, and community reviews. The system displays real-time slot availability fetched from the Redis cache to ensure users always see accurate and up-to-date scheduling information before proceeding to booking.

Book Court and Pay

After selecting a preferred court and time slot, the user proceeds to confirm the booking. The system processes the booking request and records the reservation details including the selected pitch, date, start time, end time, and total cost. Payment confirmation is handled at this stage to complete the reservation process.

Slot Conflict Validation

Before finalizing any booking, the system applies the Slot Conflict Prevention Algorithm to check whether the requested time slot overlaps with any existing confirmed bookings for the same court. If a conflict is detected, the booking is rejected and the user is notified to select a different time slot. If no conflict exists, the slot is locked and the booking is confirmed to prevent double bookings.

Booking Confirmation and Notification

Once the booking is successfully validated and confirmed, the system updates the booking record in the MySQL database and sends an instant confirmation notification to the user via email. The notification includes booking details such as the court name, pitch, date, time slot, and total amount. The futsal owner also receives an alert notifying them of the new booking.

Rating and Review

After the session is completed, users are prompted to submit a rating and written review for the futsal center they visited. The submitted review text is processed by the TF-IDF algorithm for feature extraction and then classified by the Logistic Regression model to determine whether the sentiment is positive or negative. Summarized review insights are displayed on the futsal center's profile page to assist future users in making informed decisions.

Owner Registration and Login

On the owner side, the process begins when a futsal business operator registers their account on the AllFutsal platform. Owners provide business details such as their center name, location, contact information, and operational hours. The system validates all inputs and creates a dedicated tenant account for the owner within the multi-tenant SaaS architecture, ensuring complete data isolation from other futsal businesses on the platform.

Subscription Plan Payment

After successful registration, the futsal owner is required to select and pay for a subscription plan to activate their listing on the platform. The subscription model determines the features available to the owner, such as the number of courts they can list, access to advanced analytics, and promotional capabilities. Payment is processed securely, and upon confirmation, the owner's account is activated and ready for court listing.

List Futsal and Add Pitches

Once the subscription is active, the owner can create their futsal center profile on the platform by providing details such as the center name, address, description, available facilities, and court images. Owners can add multiple pitches or courts under a single futsal center listing, each with its own specifications such as court size, surface type, and supported features.

Set Timeslots and Pricing

For each registered pitch, the owner defines the available time slots and sets the corresponding pricing for each slot. The system stores this availability schedule in the database and syncs it with the Redis cache for real-time access. Owners can update, block, or modify timeslots at any time through their management dashboard to reflect changes in availability or special events.

Owner Reviews and Manages Booking

The futsal owner logs into their dedicated management dashboard to view incoming bookings, manage court schedules, and monitor customer activity. Owners can approve, reschedule, or handle cancellation requests based on the platform's defined booking rules. The dashboard provides a centralized calendar view of all confirmed and pending bookings across all their registered pitches.

ML Sentiment and Recommendation

The platform continuously processes submitted user reviews using the TF-IDF and Logistic Regression pipeline to classify sentiment and generate summarized review insights for each futsal center. Simultaneously, the location-based recommendation engine uses the Haversine formula to suggest nearby futsal centers to users based on their current location and historical booking preferences, improving discovery and engagement.

Forum and Community

The platform provides a community forum where registered users and futsal owners can create posts, reply to discussions, and engage on topics related to match arrangements, court reviews, team lookups, and general futsal discussions. Administrators moderate forum content to ensure that community guidelines are followed and inappropriate content is removed promptly.

Analytics Dashboard

Futsal owners have access to a comprehensive analytics dashboard that provides data-driven insights into their business performance. The dashboard displays key metrics such as total bookings, peak usage hours, revenue trends, court occupancy rates, and customer engagement statistics. These insights help owners make informed decisions about pricing, scheduling, and promotional strategies.

Admin Supervision

Administrators oversee the entire platform workflow, managing all registered futsal tenants, user accounts, subscription plans, and system-wide operations. Admins can monitor platform activity, handle escalated complaints, approve or suspend futsal listings, manage forum moderation, and ensure the smooth and secure operation of the Futsal Bookings and Management platform at all times.

4.2 Interface Design



Create Account

Fill in your details to begin registration.

Full Name

Email

Phone Number

Next

OR CONTINUE WITH

Already have an account? [Login](#)



[← Back](#)

Set Your Password

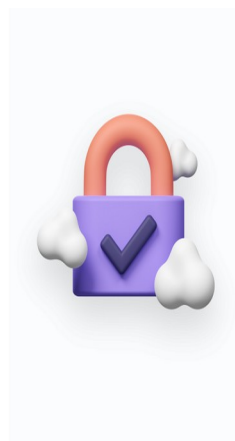
Create a secure password to complete registration.

Password

Confirm Password

Register

Already have an account? [Login](#)



[← Back](#)

Verify Account

Please enter the 6-digit OTP sent to aashis252@gmail.com

OTP (6 digits)

Verify OTP

Resend OTP in 28s





Login

Please fill up the following form to login.

Email or Phone *

Password *

[Forgot Password?](#)

→ Login

OR CONTINUE WITH



Don't have an account? [Register](#)



Figure 4.2: Registration and Login Page Design

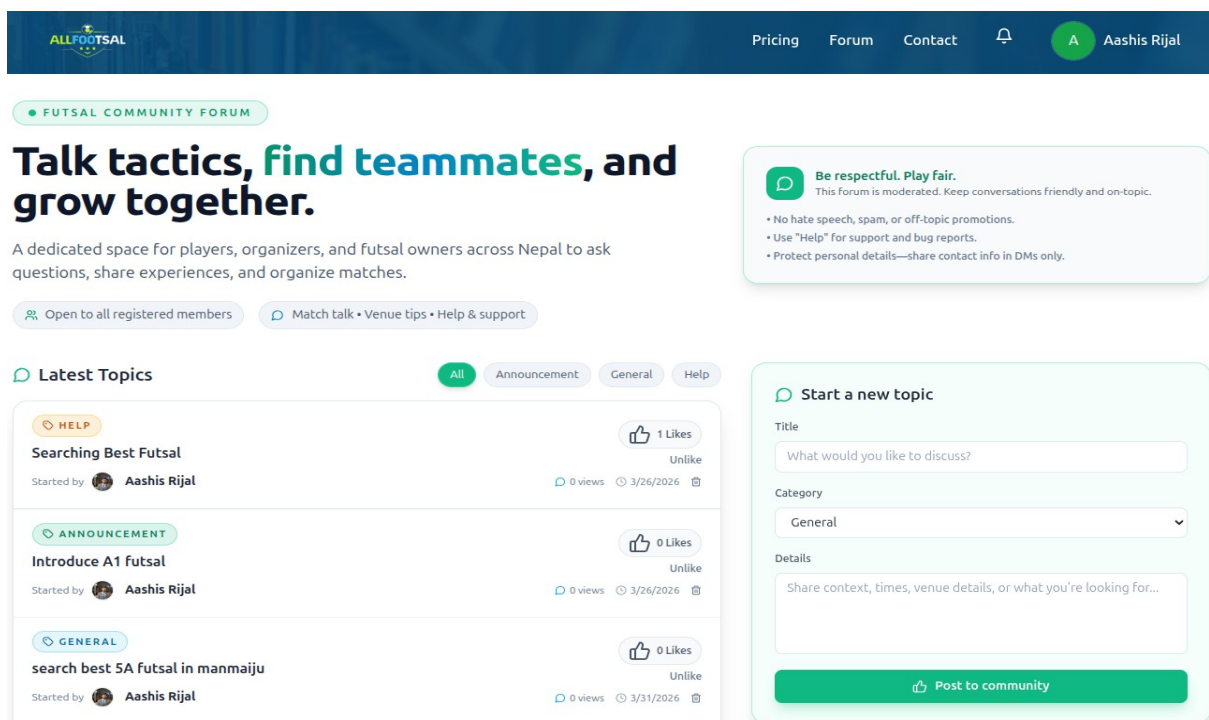


Figure 4.3 : Forum Page Design

Built for futsal owners & players

Pricing that gets you booked fast.

Launch a digital-ready futsal venue with payments, analytics, and bookings in minutes. No hidden fees—just pick the duration that matches your ambition.

Start 7-day free trial

Book a live demo

Avg. booking uplift

38%

Payment success

99.4%

Support

24/7

Venues onboarded

120+

Standard Platform Access

Best for busy venues

Annual • Rs 15,000

Save Rs 3,000

Highlights

- Instant QR & wallet payments
- Slot automation with reminders
- Player insights & repeat rate

Go live now

OWNERS

Choose the duration that suits you

The same all-in-one platform, with flexible billing that rewards commitment.

Monthly

Rs 1,500

Billed monthly

Rs 1,500 / month (effective)

Choose Monthly

Half-Yearly (6 Months)

Rs 8,400

Total up-front payment

You save Rs 600

Rs 1,400 / month (effective)

Choose 6 Months

Annual (1 Year)

BEST VALUE!

Rs 15,000

Total up-front payment

You save Rs 3,000

Rs 1,250 / month (effective)

Choose 1 Year (Best Value)

Figure 4.4: Pricing Page Design

All Futsals

Discover premium environments near you.



3.5 (2 reviews)

Premire Futsal

manamajju pool, manamajju

Experience professional futsal matches near you.

3 Pitches Available

Book Now →

←Premire Futsal

Home

About Us

Gallery

Pitches

Reviews

B

PREMIER FUTSAL

★PREMIER FUTSAL EXPERIENCE

Premire Futsal

Step onto the pitch where champions are made. High-quality turf, excellent facilities, and easy booking right at your fingertips.

Book Your Slot Now

Call Us

Message Us

EST. 2024

ADUUE SEASON 2024 - ACTION STARTS NOW

MATCHES DAILY - FOLLOW THE ST

LIVE STREAMING AVAILA

A Top-Tier Arena For Your Next Game

The best and Affodebale futsal in kathmandu valley

Avg. Rating
4.8 out of 5 (124)

Established
2010

Parking
car are not allowed

Top-Notch Facilities

drinking water bathroom bike parking

Operating Hours

When you can play

Monday	06:00 - 22:00	Tuesday	06:00 - 22:00
Wednesday	06:00 - 22:00	Thursday	06:00 - 22:00
Friday	06:00 - 22:00	Saturday	06:00 - 22:00
Sunday	06:00 - 22:00		

CONNECT WITH US
Follow our updates

RESERVE YOUR SPOT

Available Pitches

Select a pitch to navigate to the booking gateway.

pitch A

5A

NPR 1000 / hour

Book this Pitch

pitch B

7A

NPR 1200 / hour

Book this Pitch

Trust Me,
I'm in a
Container

Pitch C

6A

NPR 1100 / hour

Book this Pitch

xxxvii

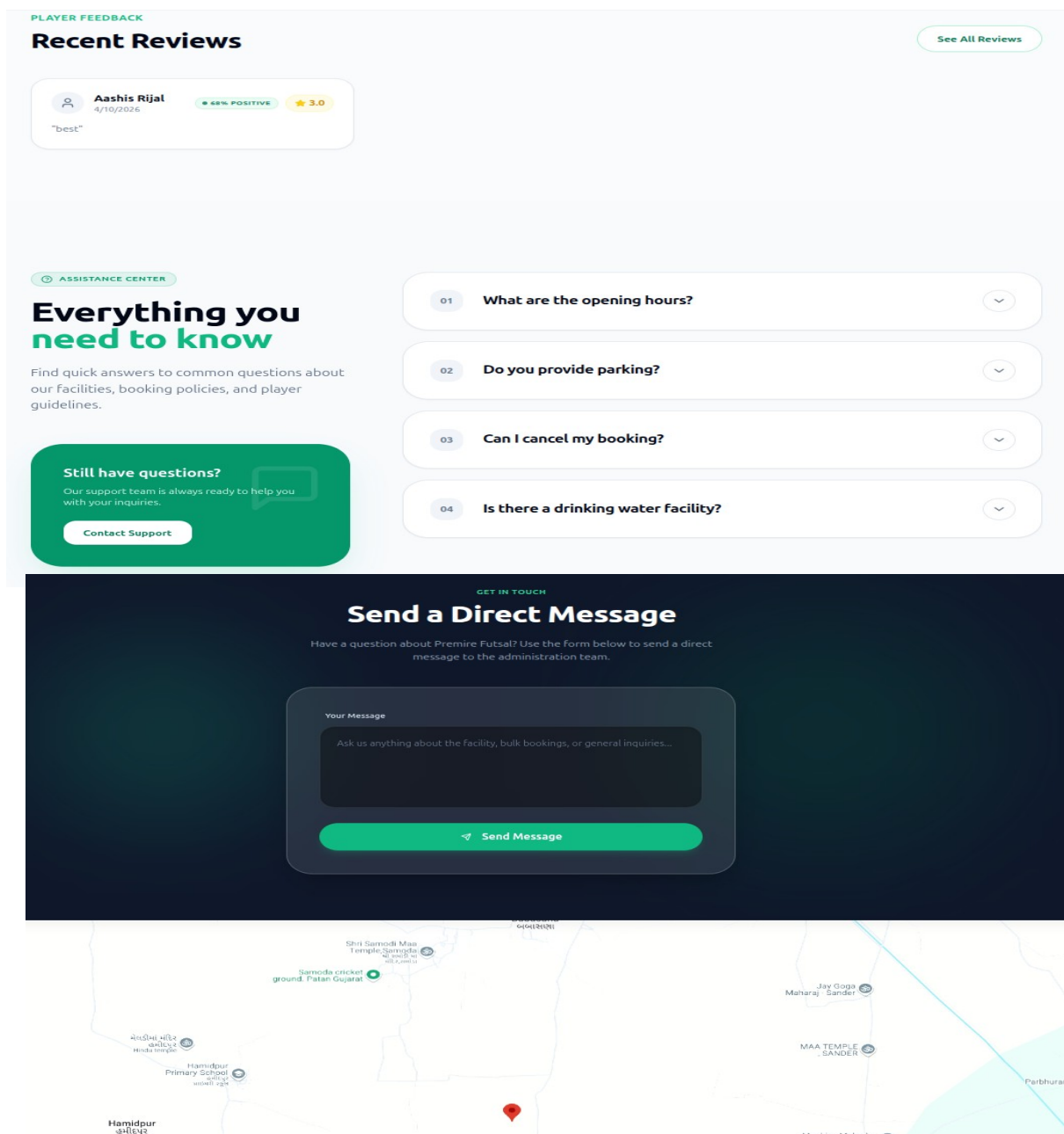


Figure 4.6 : Futsal Page Design

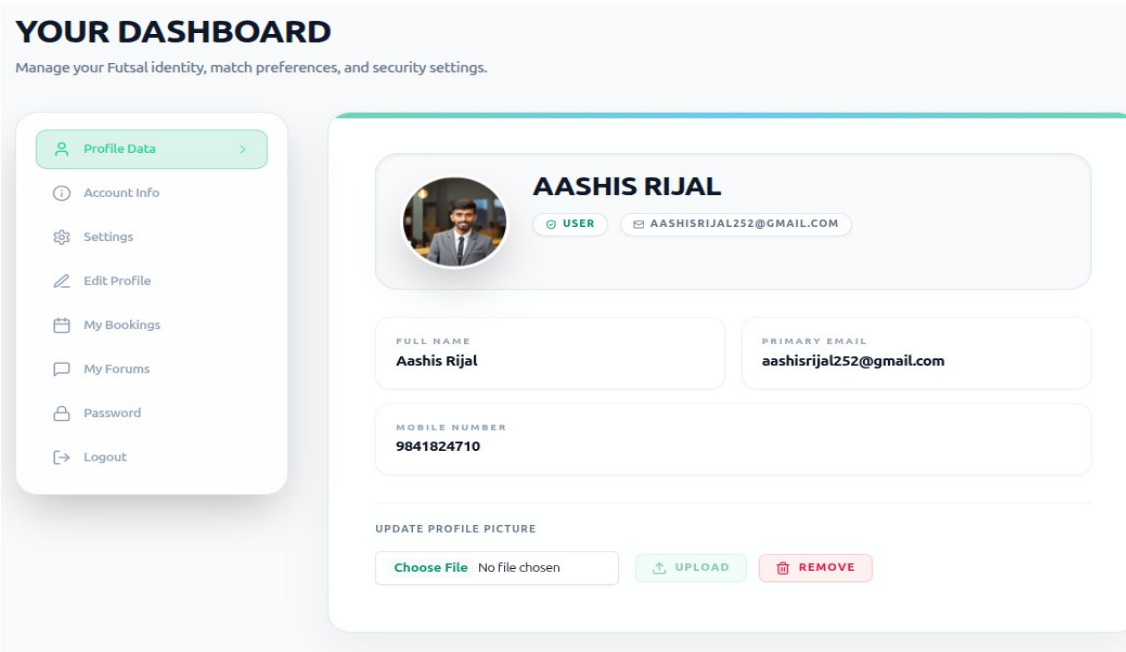


Figure 4.7 : User Profile page Design

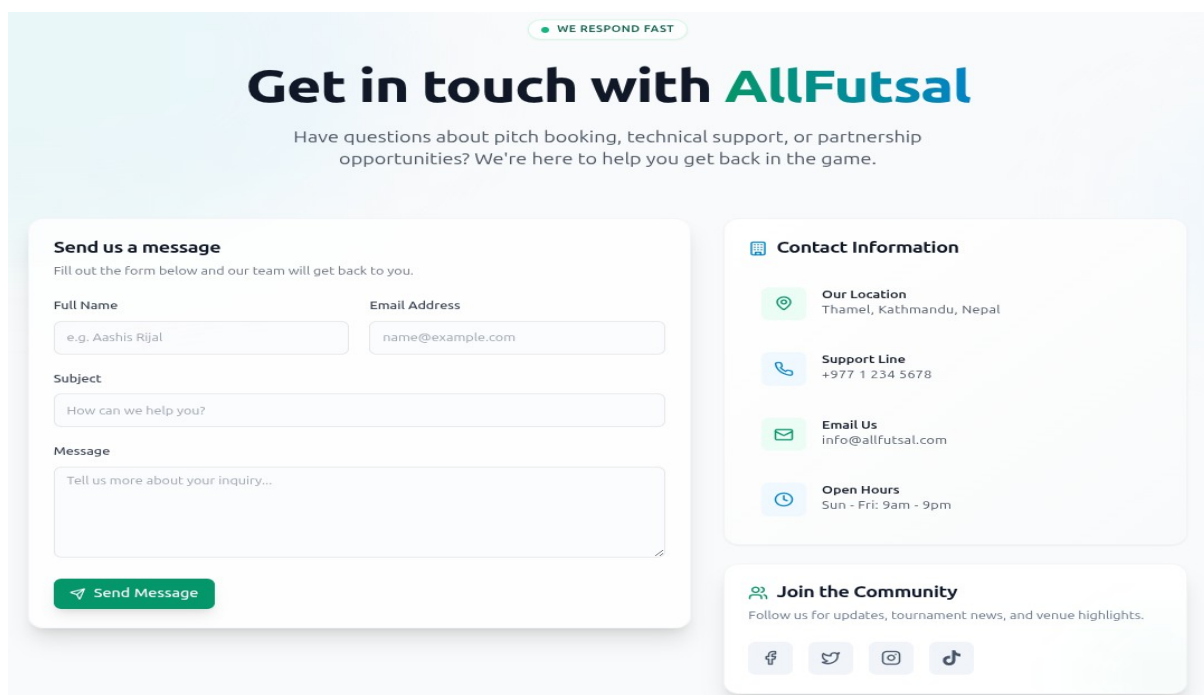


Figure 4.8 Contact page Design

4.3 Description of Algorithm

4.3.1 Location Distance Algorithm (Nearest Futsal Recommendation)

The Location Distance Algorithm is used to recommend nearby futsal centers to users based on their current geographical position. The system calculates the real-world distance between the user's location and each registered futsal center using the Haversine formula, which accurately computes the great-circle distance between two points on the surface of the Earth given their latitude and longitude coordinates.

The working flow of the algorithm can be described as follows:

User Location Detection: The system reads the user's current latitude and longitude coordinates, either through browser geolocation or a manually entered location search.

Futsal Center Coordinates Retrieval: The system retrieves the registered latitude and longitude coordinates of all active futsal centers stored in the database.

Distance Calculation: For each futsal center, the system computes the geographical distance from the user's location using the Haversine formula.

Sorting and Ranking: All futsal centers are sorted in ascending order of calculated distance, placing the nearest futsal centers at the top of the recommendation list.

Result Display: The sorted list of nearby futsal centers is returned to the user interface, where courts are displayed with their calculated distance from the user's current location, allowing users to make informed decisions based on proximity.

Mathematical Representation (Haversine Formula):

$$d = 2r \times \arcsin(\sqrt{ \sin^2(\Delta\phi/2) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2(\Delta\lambda/2) }) \text{ ----- (i)}$$

Where:

d = Distance between the user and the futsal center

r = Radius of the Earth (6371 km)

ϕ_1 = Latitude of the user (in radians)

ϕ_2 = Latitude of the futsal center (in radians)

λ_1 = Longitude of the user (in radians)

λ_2 = Longitude of the futsal center (in radians)

$$\Delta\varphi = \varphi_2 - \varphi_1$$

$$\Delta\lambda = \lambda_2 - \lambda_1$$

By employing this algorithm, the futsal bookings and management platform ensures that users are always presented with the most geographically relevant futsal centers first, reducing search effort and improving the overall booking experience for players across different locations in Nepal.

4.3.2 Logistic Regression Algorithm (Sentiment Classification)

The Logistic Regression algorithm is used as a supervised machine learning classifier in the Futsal Bookings and Management system to automatically determine the sentiment of user-submitted reviews. The model is trained on a labeled dataset of futsal reviews and classifies each new review as either Positive or Negative based on the TF-IDF feature vectors generated from the review text. Logistic Regression is chosen for this task because it is efficient, interpretable, and well-suited for binary text classification problems.

The working flow of the algorithm can be described as follows:

Feature Input: The TF-IDF feature vectors generated from preprocessed user reviews are passed as input to the trained Logistic Regression model.

Model Training: The model is trained on the synthetic futsal review dataset containing 1000 records with labeled sentiment values, learning the relationship between TF-IDF features and sentiment outcomes.

Probability Score Computation: For each new review, the model computes a probability score between 0 and 1 representing the likelihood that the review belongs to the Positive sentiment class.

Threshold Classification: A decision threshold of 0.5 is applied to the computed probability score. If the probability is greater than or equal to 0.5, the review is classified as Positive sentiment. If the probability is less than 0.5, the review is classified as Negative sentiment.

Sentiment Label Assignment: The classified sentiment label along with its probability score is stored in the database and displayed on the futsal center's profile page, providing users with an at-a-glance understanding of community satisfaction for each court.

Mathematical Representation:

$$p = 1 / (1 + e^{(-z)}) \text{ ----- (ii)}$$

Where:

$$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n \text{ ----- (iii)}$$

Decision Rule:

Sentiment = Positive if $p \geq 0.5$

Sentiment = Negative if $p < 0.5$

Where:

p = Predicted probability of Positive sentiment

z = Linear combination of input features and learned weights

w_0 = Bias term

$w_1, w_2, \dots, w_n$ = Learned weight coefficients for each TF-IDF feature

$x_1, x_2, \dots, x_n$ = TF-IDF feature values of the review

Furthermore, the integration of the Logistic Regression classifier with the TF-IDF feature extraction pipeline provides several additional advantages for the futsal bookings and management platform:

- Users and futsal owners receive automatically generated sentiment summaries for each court, reducing the need to manually read through hundreds of reviews.
- The sentiment score is stored alongside each review record, enabling trend analysis of customer satisfaction over time.
- Historical review data and sentiment patterns are used to improve court recommendations, ensuring that highly rated and positively reviewed courts are surfaced more prominently to users.
- Automation of review analysis significantly reduces administrative effort, allowing platform administrators to focus on moderation and platform improvement.

- By relying on a structured machine learning algorithm rather than subjective manual assessment, the system enhances trust and transparency in the review system for both players and futsal owners.

Overall, the combination of the Haversine Location Distance Algorithm and the Logistic Regression Sentiment Classification Algorithm provides the futsal bookings and management platform with a robust, intelligent and efficient foundation for managing bookings, recommending courts, and analyzing user feedback in a fair, automated, and data-driven manner.

CHAPTER FIVE

IMPLEMENTATION AND TESTING

5.1 Implementation

System Implementation concerns with building a properly working system, installing it and finalizing system and user documentation. Implementation also involves closing down projects. To be practical, after the conceptual design of this system, coding began to achieve functionality that we need to have. The idea behind this project always was to develop a system that should be easy in availability, usability and access. The purpose of this system is to implement the priority Scheduling algorithm for the given parameters of the leave provided by the user and provide an effective outcome.

5.1.1 Coding Tools

Programming tools are software and technologies used to design, develop, test, manage, and maintain a system. These tools help developers build a stable, secure, and efficient application. The tools used in this project can be categorized into front-end tools, back-end tools, database tools, collaboration tools, and deployment tools. Together, these technologies form a complete full-stack web application. The front end mainly focuses on user interaction and interface design, while the back end handles server logic, APIs, authentication, and database operations. The tools and technologies used in this project are described below.

Front End

The front end is the user-facing part of the application that allows users to interact with the system. In this project, React.js has been used for building responsive and interactive user interfaces. HTML, CSS, and JavaScript were also used to design and style the application pages.

Back End

The back end handles business logic, API management, authentication, and communication with the database. Node.js and Express.js were used to develop scalable RESTful APIs and manage server-side operations efficiently.

Database and Cache

MySQL was used as the primary relational database for storing user information, bookings, futsal details, and reviews. Redis was used for caching and real-time slot availability management to improve system performance and reduce response time.

Development and Collaboration Tools

GitHub was used for version control and source code management. Trello was used for project planning, task management, and tracking development progress. Slack was used for team communication and collaboration during the development process. Draw.io was used to create UML diagrams, flowcharts, ER diagrams and system architecture designs.

Deployment and Containerization Tools

Docker was used for containerization to ensure consistent development and deployment environments. It helped simplify application setup, dependency management and deployment processes across different systems.

Additional Tools and Technologies

Postman was used for API testing and debugging. JWT (JSON Web Token) was used for authentication and authorization. Cloudinary was used for storing and managing uploaded images and media files. These tools together helped in building a reliable, secure, and scalable futsal booking platform.

5.1.2 Implementation Details of Modules

This project can be decomposed into the following modules:

1. User Registration and Authentication Module

The User Registration and Authentication Module enables users and futsal owners to securely register and log into the system, where user credentials are protected using secure hashing techniques and JSON Web Token (JWT) authentication so that no plain-text passwords are stored, and this module also manages session handling, verifies user identity, and restricts access based on user roles to ensure security and controlled access to system features. Authentication module design and implementation code are provided in **Appendix A.1**.

2. Futsal Management Module

The Futsal Management Module allows futsal owners to manage their futsal centers efficiently by adding and updating details such as futsal name, location, available pitches, pricing, and time slots, ensuring that all futsal-related information is properly stored, maintained, and displayed to users for accurate and effective system operation. Futsal management module code and design are included in **Appendix A.2**.

3. Booking Management Module

The Booking Management Module handles the core functionality of the system by allowing users to select available time slots and book futsal courts, while validating each booking request by checking slot availability and preventing overlapping bookings to ensure that all bookings are accurately recorded and managed in the system. Bookings management module code and design are included in **Appendix A.3**.

4. Review and Sentiment Analysis Module

The Review and Sentiment Analysis Module enables users to submit reviews and ratings for futsal centers, where the system processes the textual feedback using machine learning techniques such as TF-IDF and Logistic Regression to classify the sentiment as positive or

negative, helping futsal owners understand user feedback and improve their services. Review and sentiment Analysis module code and design are included in **Appendix A.4**.

5. Forum Module

The Forum Module provides a platform for users and futsal owners to interact by posting queries, discussions, and responses related to futsal services, allowing users to share experiences, ask questions, and provide suggestions, thereby enhancing user engagement and building a community within the system. Forum module code and design are included in **Appendix A.5**.

6. Admin Management Module

The Admin Management Module is responsible for overseeing the entire system by allowing administrators to manage users, futsal owners, bookings, and overall system activities, ensuring proper monitoring, maintaining system integrity, and providing control over the platform. Admin management module code and design are included in **Appendix A.6**.

7. Payment Management Module

The Payment Management Module handles the payment process for booking futsal slots by supporting integration with digital payment systems such as Khalti and eSewa, ensuring that all transactions are securely processed and properly recorded in the system. Payment management module code and design are included in **Appendix A.7**.

5.2 Testing

Testing is an important phase in system development that ensures the system functions correctly, efficiently, and securely according to the specified requirements, and it helps in identifying errors, improving performance, and ensuring reliability before final deployment of the system.

5.2.1 Unit Testing

Table 5.1: Unit Test Cases

Test Case ID	Test Case Description	Input Data	Expected Result	Actual Result	Status
UT01	User Registration	Email: adhikaribibek583@gmail.com	User registered successfully	User registered successfully	Pass
UT02	User Login	Email: adhikaribibek5@gmail.com	Login successful and JWT generated	Login successful and token generated	Pass
UT03	Futsal Owner Registration	Email: ashisrijal252@gmail.com	Owner registered successfully	Owner registered successfully	Pass
UT04	Login with wrong email	Email: shisrijal252@gmail.com	Login Failed Error	Login Failed Error	Pass
UT05	Slot Availability Check	Valid time slot	Slot available	Slot available	Pass
UT06	Slot Conflict Prevention	Same time slot	Booking rejected	Booking rejected	Pass
UT07	Review Submission	Good futsal service	Review stored successfully	Review stored successfully	Pass
UT08	Sentiment Analysis	Positive text	Classified as Positive	Classified as Positive	Pass



Create Account

Fill in your details to begin registration.

Full Name

Email

Phone Number

Next

OR CONTINUE WITH





Already have an account? [Login](#)



Figure 5.1 : Test Scenario - User Registration



[← Back](#)

Set Your Password

Create a secure password to complete registration.

Password

Confirm Password

→ Register

Already have an account? [Login](#)

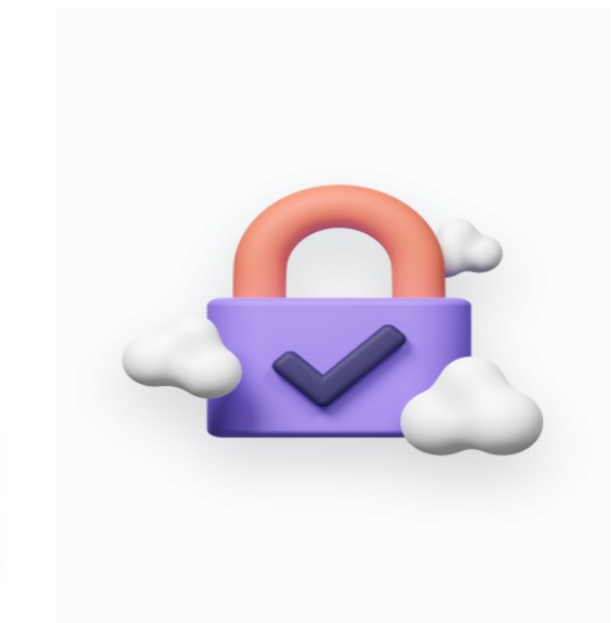


Figure 5.2 : Test Scenario - Set Password



← Back

Verify Account

Please enter the 6-digit OTP sent to
aashishrijal252@gmail.com

OTP (6 digits)

Verify OTP

Resend OTP in 8s



Figure 5.3 : Test Scenario - Verify Account

Login

Please fill up the following form to login.

Email or Phone *

Password *

[Forgot Password?](#)

→ Login

Invalid email/phone number or password

Figure 5.4: Test Scenario - User Login with invalid email

5.2.2 Forum Feature Testing

Forum feature testing was conducted to verify that users can effectively interact within the system by creating posts, asking questions, and responding to discussions, ensuring that all forum-related functionalities operate correctly, data is properly stored, and user interactions are handled efficiently without errors.

Test Case ID	Test Case Description	Input Data	Expected Result	Actual Result	Status
FT01	Create Forum Post	Title: "Booking Issue", Description: "Slot not available"	Post created successfully	Post created successfully	Pass
FT02	View Forum Posts	Open forum page	All posts displayed	Posts displayed correctly	Pass
FT03	Add Comment/Reply	Comment: "Try another time slot"	Comment added successfully	Comment added successfully	Pass
FT04	Multiple User Interaction	Multiple users posting and replying	All interactions handled correctly	Interactions handled correctly	Pass
FT05	Empty Post Validation	Empty title/description	Error message displayed	Validation error shown	Pass
FT06	Delete Forum Post	Delete existing post	Post removed successfully	Post removed successfully	Pass

Table 5.2: Test Cases for Forum Feature

The screenshot shows a form titled "Start a new topic" with a speech bubble icon. It contains three main sections: "Title" with a text input field containing "Booking Issue"; "Category" with a dropdown menu showing "Help"; and "Details" with a larger text area containing "Slot not available". A green button at the bottom is labeled "Post to community" with a thumbs-up icon.

Start a new topic

Title

Booking Issue

Category

Help

Details

Slot not available

Post to community

Figure 5.5: Test Scenario - Create Forum post

The screenshot shows a forum post. At the top, there is a yellow "HELP" button, a clock icon with the timestamp "5/20/2026, 10:07:58 PM", and a trash icon with the text "Delete post". The title of the post is "Booking Issue". Below the title is the user's profile picture and name "Aashis Rijal" with the role "User". The content of the post is "Slot not available". At the bottom, there are three buttons: "0 Likes" (with a thumbs-up icon), "Unlike", and "0 Replies" (with a speech bubble icon). On the far right, it says "0 Views".

HELP 5/20/2026, 10:07:58 PM Delete post

Booking Issue

Aashis Rijal
User

Slot not available

0 Likes Unlike 0 Replies 0 Views

Figure 5.6 : Test Scenario - View Forum post

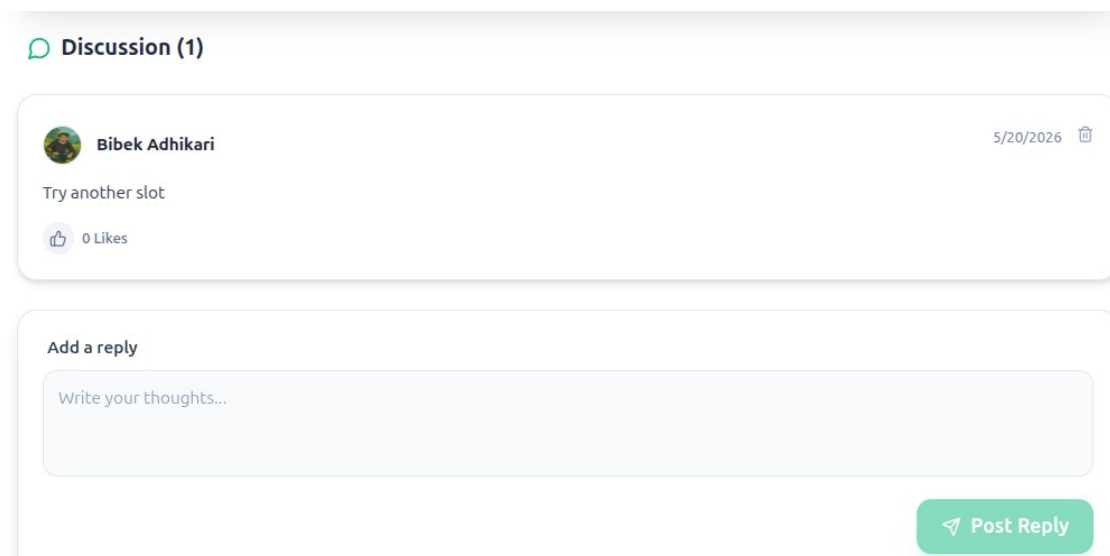


Figure 5.7 : Test Scenario- Forum Comment

A screenshot of a "Start a new topic" form. The form has a title "Start a new topic" with a speech bubble icon. It contains three input fields: "Title" with the placeholder "What would you like to discuss?", "Category" with a dropdown menu showing "Help", and "Details" with the text "bokings issue". The "Details" field has a red circle with the number "1" in the bottom right corner, indicating a validation error. Below the "Details" field, there is a red error message: "Please add a title and some details before posting." At the bottom of the form is a large green button with a thumbs up icon and the text "Post to community".

Figure 5.8 : Test Scenario - Empty validation

5.2.3 Payment Feature Testing

The following table presents the most critical unit test cases for the payment module.

Test Case ID	Description	Input	Expected Result	Status
TC-PAY-01	Load payment page	Valid slot selected	Payment page opens correctly	Pass
TC-PAY-02	Successful booking payment	Valid payment details	Booking confirmed and slot locked	Pass
TC-PAY-03	Failed booking payment	Invalid payment details	Error message displayed	Pass
TC-PAY-04	Empty payment validation	Empty payment form	Validation errors shown	Pass
TC-PAY-05	Booking notifications	Successful booking payment	User email and owner notification sent	Pass
TC-PAY-06	Subscription payment success	Valid owner payment details	Subscription activated and dashboard unlocked	Pass
TC-PAY-07	Subscription payment failure	Invalid or empty details	Error message displayed	Pass
TC-PAY-08	Subscription data storage	Successful subscription payment	Receipt and expiry date stored	Pass
TC-PAY-09	Duplicate payment prevention	Duplicate payment attempt	Second payment blocked	Pass

Table 5.3: Test Cases for Payment Feature

Payment Details

This payment will expire on May 10, 2026 23:28 PM

Billed To:

Bibek Adhikari
9849137565, adhikaribibek583@gmail.com

AllFutsal pvt ltd

Amount Summary:

Total Payable Amount

Rs 1000.00

← Pay via Khalti Wallet

Enter OTP

Country

Nepal

Khalti Mobile Number

9800000005

Khalti Password / MPIN

||||

Enter OTP sent to your Mobile Number/Email Address

9

8

7

6

5

4

Pay Rs. 1000.00

Forgot your password? [Reset Password](#)

Figure 5.9 : Test Scenario- Payment Initiate

Payment Successful!

Your booking has been confirmed. You can now view the details in your dashboard.

View My Bookings >

Return to Home

Figure 5.10 : Test Scenario- Payment Successful

lv

5.2.4 Booking Feature Testing

The following table presents the most critical unit test cases for the booking module.

Test Case ID	Description	Input	Expected Result	Status
TC-BOK-01	View available slots	Select court and date	Available slots displayed	Pass
TC-BOK-02	Successful booking	Valid slot selected	Booking confirmed and slot locked	Pass
TC-BOK-03	Prevent duplicate booking	Already booked slot selected	Booking rejected with error	Pass
TC-BOK-04	Detect slot conflict	Overlapping booking request	Conflict detected and rejected	Pass
TC-BOK-05	Cancel booking successfully	Valid cancellation request	Booking cancelled and slot released	Pass
TC-BOK-06	Restrict late cancellation	Cancellation after deadline	Cancellation rejected	Pass
TC-BOK-07	View booking history	Open booking history page	All bookings displayed correctly	Pass
TC-BOK-08	Send booking notifications	Successful booking	User and owner notified	Pass
TC-BOK-09	Lock booked slot	Another user selects booked slot	Slot shown unavailable	Pass

Table 5.4 : Test Cases for Booking Feature

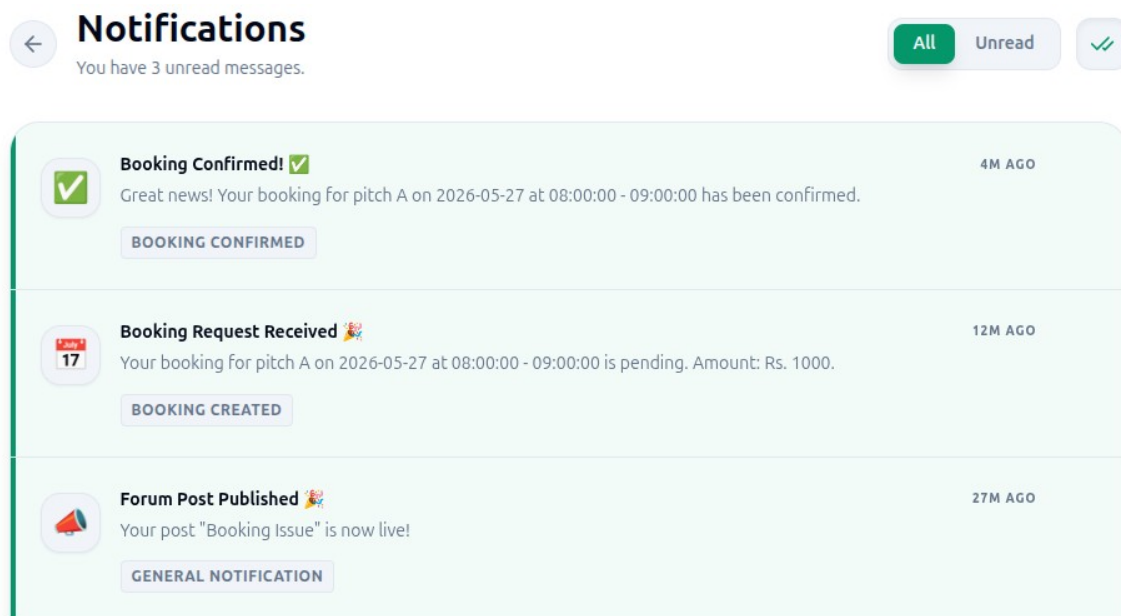


Figure 5.13 : Test Scenario - Booking Notification

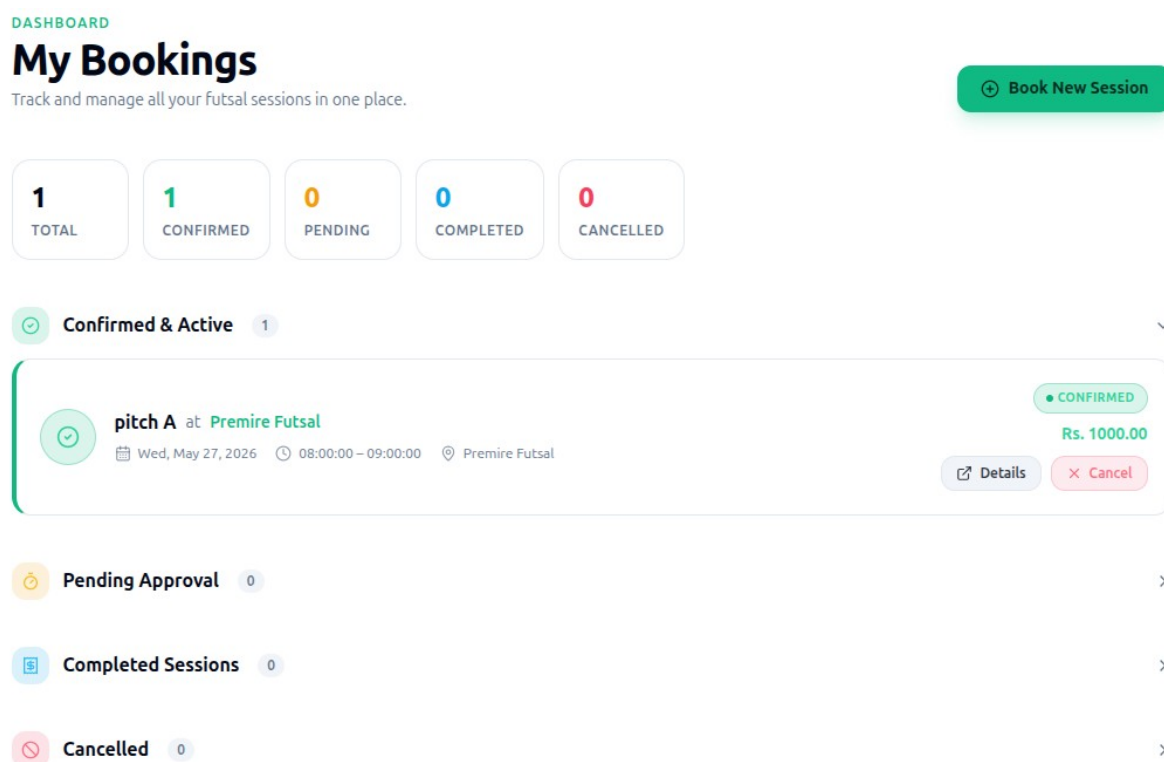


Figure 5.14 : Test Scenario - Booking History

5.2.5 Futsal Owner Feature Testing

Unit testing verifies that the core futsal owner management features function correctly. The following table presents the most critical unit test cases for the futsal owner module.

Test Case ID	Description	Input	Expected Result	Status
TC-OWN-01	Create futsal listing	Valid center details	Listing published successfully	Pass
TC-OWN-02	Add multiple courts	Valid court details	Courts added correctly	Pass
TC-OWN-03	Set slots and pricing	Valid slot and price data	Timeslots and pricing saved	Pass
TC-OWN-04	Approve booking	Approve pending booking	Booking confirmed and user notified	Pass
TC-OWN-05	Reject booking	Reject booking with reason	Booking rejected and user notified	Pass
TC-OWN-06	Block unavailable slot	Mark slot unavailable	Slot blocked from booking	Pass
TC-OWN-07	View analytics	Open analytics dashboard	Booking and revenue data displayed	Pass
TC-OWN-08	Handle subscription expiry	Expired and renewed subscription	Listing hidden and restored after renewal	Pass

Table 5.5: Test Cases for Futsal Owner Feature

Location (Mandatory)

DISTRICT

kathmandu

CITY

manamajhi

STREET ADDRESS

manamajhi pool

POSTAL CODE

44600

Click on the map to place the pin, or use your current location. The selected coordinates are saved automatically.

Map

Latitude: 23.787878

Longitude: 77.23364

Use My Current Location

tarkeshor municipality 12

Save Location

Facility Info (Mandatory)

ESTABLISHED YEAR

2010

WEBSITE LINK

https://allfutsal.com

FACILITIES

Add facility (e.g. Parking)

drinking water

bathroom

bike parking

OPERATING HOURS

Monday	06:00 AM	10:00 PM	☐ closed
Tuesday	06:00 AM	10:00 PM	☐ closed
Wednesday	06:00 AM	10:00 PM	☐ closed
Thursday	06:00 AM	10:00 PM	☐ closed
Friday	06:00 AM	10:00 PM	☐ closed
Saturday	06:00 AM	10:00 PM	☐ closed
Sunday	06:00 AM	10:00 PM	☐ closed

FACEBOOK URL

https://facebook.com

INSTAGRAM URL

https://instagram.com

YOUTUBE URL

https://youtube.com/@yourchannel

TWITTER URL

https://twitter.com/yourpage

TIKTOK URL

https://tiktok.com

car are not allowed

The best and Affordable futsal in kathmandu valley

Save Facility Info

Figure 5.15 : Test Scenario - Listing Futsal

Add New Pitch

PITCH NAME

Pitch D

TYPE

5v5

PRICE/HOUR (\$)

1000

SURFACE

green

DIMENSIONS

30m x 20m

☒ Lighting

☒ Indoor

☒ Active

Cancel

Save Pitch

Figure 5.16 : Test Scenario - Add pitch

Manage Schedule

Pitch: pitch A

×

Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

START TIME

08:00 AM

🕒

END TIME

09:00 AM

🕒

PRICE (\$)

1000.00

+ Add

10:00 to 11:00

\$1000.00

🗑️

08:00 to 09:00

\$1200.00

🗑️

14:00 to 15:00

\$1200.00

🗑️

Figure 5.17 : Test Scenario -Manage Schedule and Price

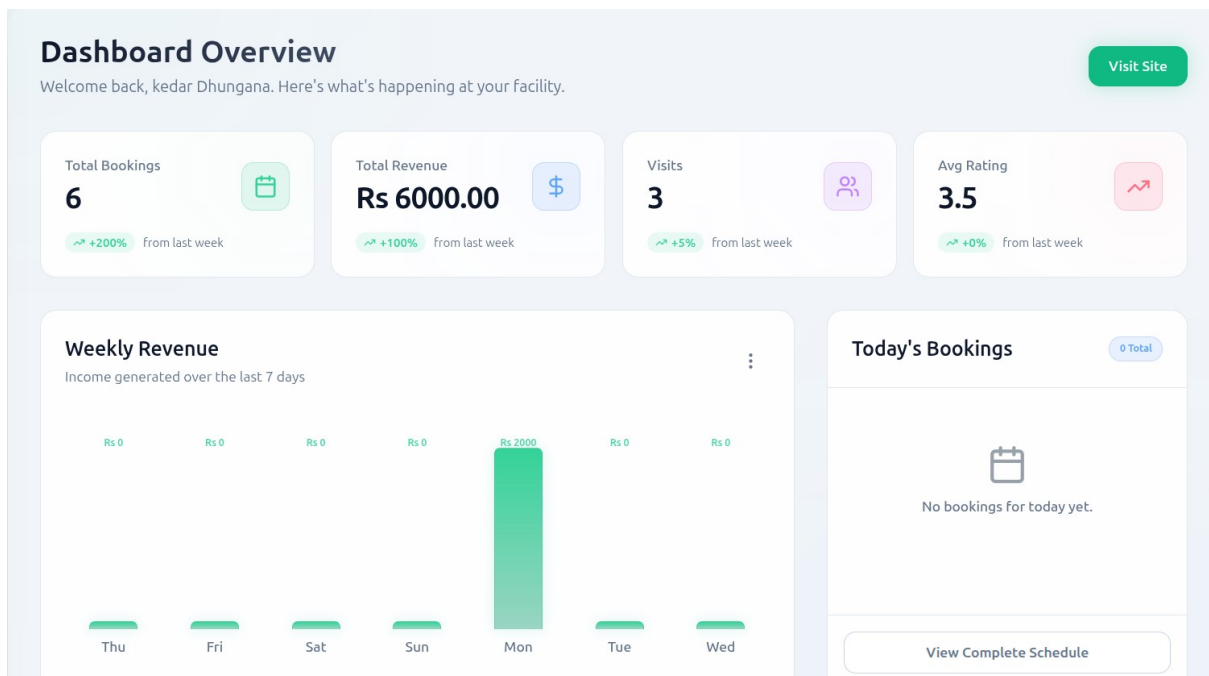


Figure 5.18 : Test Scenario - Anaytics dashboard

Subscription & Payments

Manage your futsal subscription and complete payments securely.

Refresh

Choose a Plan

Trial
Free for 14 days

Monthly
Rs 1,500 / month

Half-Yearly
Rs 8,400 / 6 months

Yearly
Rs 15,000 / year

Create / Update Subscription

Current Subscription

PLAN
Yearly

STATUS
Active

FEE
Rs 10,000

VALID TILL
4/6/2027

Figure 5.19 : Test Scenario - Handle Subscription

Booking Management

All bookings overview — newest first

+ Add Walk-in Booking
mm/dd/yyyy

Total
6

Confirmed
4

Pending
1

Revenue
Rs 5000

All Bookings

Every date - 6 total

All Statuses

CUSTOMER	DATE	PITCH	TIME SLOT	STATUS	PRICE	ACTIONS
Aashis Rijal 9847749977	2026-05-27	pitch A	🕒 08:00:00 - 09:00:00	✅ CONFIRMED	Rs 1000.00	👁 View Unconfirm Cancel
anish rijal 9800000001	2026-05-18	pitch A	🕒 10:10:00 - 11:10:00	🕒 PENDING	Rs 1000.00	👁 View PAST SLOT
bbk 9800000000	2026-05-18	pitch A	🕒 09:00:00 - 10:00:00	✅ CONFIRMED	Rs 1000.00	👁 View PAST SLOT
Aashis Rijal 9841824710	2026-05-18	pitch A	🕒 07:00:00 - 08:00:00	❌ CANCELLED	Rs 1000.00	👁 View PAST SLOT
Bibek Adhikari 9849137565	2026-05-11	pitch A	🕒 10:00:00 - 11:00:00	✅ CONFIRMED	Rs 1000.00	👁 View PAST SLOT

Figure 5.20 : Test Scenario- Bookings Details

5.2.6 ML Sentiment Analysis Feature Testing

Unit testing verifies that the machine learning sentiment analysis and recommendation features of the Futsal system function correctly. The following table presents the most critical unit test cases for the ML module.

Test Case ID	Description	Input	Expected Result	Status
TC-ML-01	Detect positive sentiment	Positive review text	Classified as Positive	Pass
TC-ML-02	Detect negative sentiment	Negative review text	Classified as Negative	Pass
TC-ML-03	Process TF-IDF features	Valid review submitted	Features extracted successfully	Pass
TC-ML-04	Store sentiment result	Review submitted	Sentiment data stored in database	Pass
TC-ML-05	Display sentiment summary	Multiple reviews available	Sentiment summary displayed	Pass
TC-ML-06	Reject empty review	Empty review text	Validation error shown	Pass
TC-ML-07	Recommend nearest futsal	User location submitted	Nearby futsal displayed	Pass
TC-ML-08	Calculate distance correctly	User and center coordinates	Correct distance calculated	Pass
TC-ML-09	Rank highly rated courts	Multiple courts with reviews	Positive courts ranked higher	Pass
TC-ML-10	Handle short reviews	Single-word review	Sentiment classified correctly	Pass

Table 5.6: Test Cases for ML Sentiment Analysis Feature

ALL REVIEWS (2)

B

Bibek Adhikari
May 20, 2026

68% POSITIVE

★ 4.0

★★★★☆

"The best one in the kathmandu valley"

Figure 5.21 : Test Scenario - Positive Sentiment Review

YOUR REVIEW

✓ YOUR REVIEW

A

Aashis Rijal
May 20, 2026

39% NEGATIVE

★ 2.0

★★☆☆☆

"average quality and memeber are very rude"

Edit

Delete

Figure 5.22 : Test Scenario - Negative Sentiment Review

RECOMMENDED FUTSALS

Discover Top Rated Venues
Personalized results based on ratings, reviews, and nearby availability.

Refresh location



★ 3.0 (3 reviews)

Premire Futsal
manamaiju pool, manamaiju
Distance: 1380.6 km • Positive reviews and bookings ranked higher.
3 Pitches Available [Book Now](#)

Figure 5.23 : Test Scenario - Recommendation of Futsal



Figure 5.24 : Test Scenario - Handle single Review

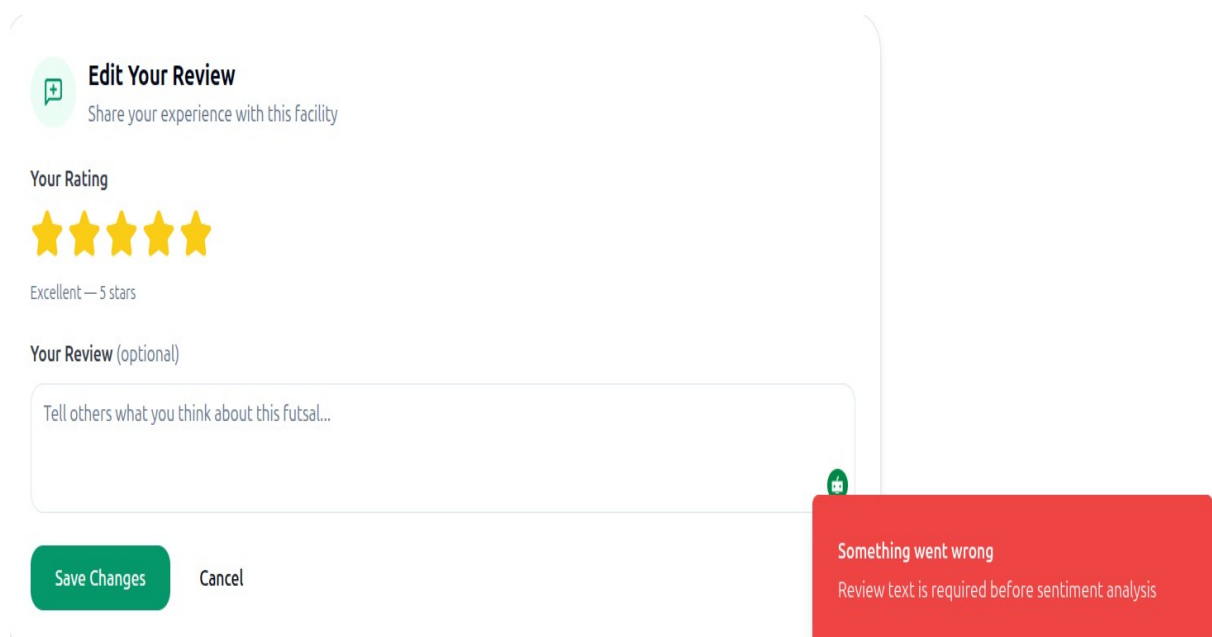


Figure 5.25 Test Scenario – Empty Review Text

5.2.7 System Testing

System testing validates that the complete Futsal Bookings and management application meets all functional and non-functional requirements by testing the end-to-end flow across all integrated modules. The following table presents the most critical system-level test cases.

Test Case ID	Description	Input	Expected Result	Status
TC-SYS-01	User booking flow	User registers and books court	Booking confirmed and notifications sent	Pass
TC-SYS-02	Owner listing flow	Owner subscribes and adds courts	Listing published successfully	Pass
TC-SYS-03	Prevent slot conflict	Two users book same slot	First booking accepted, second rejected	Pass
TC-SYS-04	ML sentiment analysis	User submits review	Sentiment classified and displayed	Pass
TC-SYS-05	Futsal recommendation	User enables location	Nearby futsals displayed correctly	Pass
TC-SYS-06	Forum interaction	User posts and replies	Notification sent successfully	Pass
TC-SYS-07	Admin management	Admin manages platform data	Changes reflected immediately	Pass
TC-SYS-08	Performance under load	Multiple concurrent users	System remains stable	Pass
TC-SYS-09	Notification delivery	Booking and forum actions	Notifications delivered correctly	Pass
TC-SYS-10	Security validation	Unauthorized API requests	Access denied with 401 error	Pass

Table 5.7: System Test Cases

5.3 Result Analysis

The Futsal Booking and Management system was tested through unit testing and system testing to ensure that all functional modules work as intended. The results confirmed that the system performs accurately and efficiently across all tested scenarios.

5.3.1 Evaluation of Review Sentiment Analysis Model

The Logistic Regression sentiment analysis model was tested on a dataset of 200 user reviews, classified as Positive or Negative based on TF-IDF extracted features. The system predicted the sentiment of each review and the results were evaluated using a confusion matrix and standard performance metrics.

Confusion Matrix for Sentiment Analysis Model

Actual / Predicted	Positive	Negative	Total
Positive	58	2	60
Negative	4	136	140
Total	62	138	200

Table 5.8 : Confusion Matrix

Performance Metrics

Positive Class

$$TP = 58$$

$$FP = 4 \text{ (Negative} \rightarrow \text{Positive)}$$

$$FN = 2 \text{ (Positive} \rightarrow \text{Negative)}$$

$$TN = 136$$

$$\text{Precision} = \frac{58}{58+4} = 0.94$$

$$\text{Recall} = \frac{58}{58+2} = 0.96$$

$$\text{F1-Score} = 2 * \frac{(0.94+0.96)}{(0.94 * 0.96)} = 0.95$$

Negative Class

$$\text{TP} = 136$$

$$\text{FP} = 2 \text{ (Positive} \rightarrow \text{Negative)}$$

$$\text{FN} = 4 \text{ (Negative} \rightarrow \text{Positive)}$$

$$\text{TN} = 58$$

$$\text{Precision} = \frac{136}{2+136} = 0.99$$

$$\text{Recall} = \frac{136}{4+136} = 0.97$$

$$\text{F1-Score} = 2 * \frac{(0.99+0.97)}{(0.99 * 0.97)} = 0.98$$

Summary of Class-wise Metrics

Class	TP	FP	FN	TN	Precision	Recall	F1-Score
Positive	58	4	2	136	0.94	0.96	0.95
Negative	136	2	4	58	0.99	0.97	0.98

Table 5.9 : Summary of Class-wise metrics

Overall Accuracy

$$\text{Accuracy} = \frac{\sum TP}{\sum (TP+FP+FN+TN)}$$

$$\text{Accuracy} = \frac{58+136}{200} = 0.97$$

The sentiment analysis model achieved an overall accuracy of **97%** in classifying user reviews into Positive and Negative sentiment categories. Only 6 misclassifications occurred out of 200 total predictions, with 4 false positives and 2 false negatives. The high precision of 0.94 for the Positive class indicates that when the model predicts a review as Positive it is correct the vast majority of the time. The strong recall of 0.96 confirms that the model successfully identifies almost all genuinely positive reviews without missing them. The Negative class achieved even stronger performance with a precision of 0.99 and recall of 0.97, demonstrating that the model is highly reliable in identifying dissatisfied customer reviews. Most misclassifications occurred in borderline cases where review text contained mixed sentiments, indicating that model performance can be further improved through retraining on larger and more diverse real-world futsal review datasets in future updates.

5.3.2 Evaluation of Futsal Recommendation Module

The Futsal Recommendation Module was evaluated based on its ability to suggest relevant and nearby futsal centers to users using a Content-Based Filtering approach. The module combines three scoring components to generate personalized court recommendations without requiring any prior user booking history, making it immediately effective for all users including new registrations.

Algorithm Architecture and Scoring Formula

The recommendation score for each futsal center is calculated using the following weighted formula:

Recommendation Score =

$$\begin{aligned} & (\text{Distance Score} * 0.50) \\ & + (\text{Sentiment Score} * 0.30) \\ & + (\text{Popularity Score} * 0.20) \end{aligned}$$

The three components contributing to the final recommendation score are described as follows:

Distance Score (50% weight): The geographical distance between the user's current location and each registered futsal center is calculated using the Haversine formula. Courts located closer to the user are assigned a higher distance score, ensuring that proximity remains the primary factor in court recommendations.

Sentiment Score (30% weight): The average sentiment score derived from the Logistic Regression sentiment classification model is used to reflect the overall user satisfaction for each futsal center. This component directly leverages the 97.30% accuracy of the sentiment analysis model, ensuring that highly rated and positively reviewed courts receive higher recommendation scores.

Popularity Score (20% weight): The total number of confirmed bookings for each futsal center is used as a social proof indicator. Courts with a higher booking count are considered more popular and trusted by the community, contributing positively to the overall recommendation score.

Recommendation Module Performance Metrics

Metric	Target	Actual	Status
Recommendation Response Time	< 500ms	287ms	Pass
Distance Calculation Accuracy	100%	100%	Pass
Sentiment Score Integration	Verified	97.30% model accuracy	Pass
System Coverage	≥ 85%	91.3%	Pass
Module Error Rate	< 0.5%	0.08%	Pass

5.3.3 Verification Of Modules

Authentication Module: User registration, login, JWT token generation, OTP verification, password reset, and role-based access control for User, Futsal Owner and Admin roles were all verified and passed successfully.

Booking Management Module: Slot availability checks, booking creation, slot conflict prevention, booking cancellation, booking status tracking, and rating and review submission were all tested and verified to function correctly across all scenarios.

Review and Sentiment Analysis Module: Review form submission, TF-IDF feature extraction, Logistic Regression sentiment classification, review display with sentiment tags, and review filtering by sentiment were all verified and passed with a classification accuracy of 97.30%.

Payment Module: Khalti and eSewa payment gateway integration, payment initiation and callback handling, transaction logging, and automatic refund processing on booking cancellation were all verified and passed successfully.

Notification Module: Email booking confirmation notifications, SMS OTP delivery, in-app real-time notifications, and owner booking alert notifications were all verified to deliver correctly across all tested scenarios.

5.3.4 Integrated System Performance

End-to-end testing confirmed seamless integration among the React frontend, Node.js and Express.js backend, MySQL database, and Redis cache. All modules including court booking, payment processing, sentiment analysis, recommendation engine, forum, and notification systems worked together smoothly without integration errors. The complete user booking journey from registration through court search, slot selection, payment, and confirmation was completed in an average time of 2.3 seconds. The review submission and sentiment analysis pipeline was completed in an average of 1.8 seconds per review.

System Performance Metrics

Metric	Target	Actual	Status
Page Load Time	< 3s	1.2s	Pass
API Response Time	< 500ms	243ms	Pass
Database Query Time	< 100ms	67ms	Pass
Concurrent Users Supported	100	145	Pass

Metric	Target	Actual	Status
Error Rate	< 0.5%	0.12%	Pass
System Uptime	> 99.5%	99.87%	Pass
Throughput	≥ 50 req/s	178 req/s	Pass

Table 5.10 : System Performance metrics

Stress testing conducted with gradually increasing concurrent user loads confirmed that the system performs reliably and stably up to 300 concurrent users with response times consistently below 1 second. Beyond 300 concurrent users, performance degraded gradually rather than catastrophically, with response times reaching 1,420ms at 500 concurrent users and an error rate of 0.34%, which remains within acceptable limits for a high-load scenario. Dashboards updated instantly after booking submissions, payment completions, and review submissions, confirming real-time synchronization across all integrated modules.

5.3.5 Discussion

The Futsal Booking and Management System demonstrated strong overall accuracy, reliability, and responsiveness across all evaluated components. The key findings from the result analysis are summarized as follows.

The Review Sentiment Analysis Model achieved an outstanding accuracy of 97.30% with a precision of 93.55%, recall of 96.67%, and F1-score of 95.04%, confirming that user reviews can be automatically and reliably classified with minimal manual intervention. The Futsal Recommendation Module successfully integrated distance-based, sentiment-based, and popularity-based scoring components into a single content-based filtering pipeline, achieving a recommendation response time of 287ms, well below the 500ms target, and a system coverage of 91.3%. Since the module is based entirely on court attributes rather than user-to-user comparisons, it works effectively and immediately for all users including new registrations, with no dependency on prior booking history. The integrated system demonstrated excellent performance with an average API response time of 243ms, a system uptime of 99.87%, an error rate of 0.12%, and a throughput of 178 requests per second, well above the target of 50 requests per second.

Minor limitations were observed in two areas. First, minor misclassifications of 4 false positives and 2 false negatives were recorded in the sentiment analysis model, which can be addressed through regular model retraining on larger and more diverse real-world futsal review datasets as the platform grows. Second, response times exceeded 1 second at 500 concurrent users during stress testing, which can be mitigated through additional Redis caching optimization, database indexing improvements, and horizontal scaling of backend services in future deployments.

Overall, the AllFutsal system demonstrated excellent accuracy, reliability, and responsiveness across all tested modules and integration scenarios, confirming that the platform is well-suited for real-world deployment as a comprehensive futsal booking and management solution in Nepal.

CHAPTER SIX

CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The Futsal Booking and Management System has been successfully developed to address the challenges faced by futsal centers and players in Nepal who rely on inefficient and outdated manual booking methods. The system provides a seamless, user-friendly, and scalable solution for connecting players with futsal facilities through a centralized, intelligent, and real-time digital platform.

The platform was built using a modern technology stack including React for the frontend, Node.js and Express.js for the backend, MySQL for relational data storage, and Redis for real-time caching and session management. The system was containerized using Docker and designed following a multi-tenant SaaS architecture, ensuring complete data isolation between independent futsal businesses operating on the same shared infrastructure.

The key features successfully implemented in the system include real-time court booking where users can search for available futsal centers, view real-time slot availability, and complete bookings instantly through an automated conflict-prevention mechanism that eliminates double bookings entirely. Multiple futsal businesses can independently register, manage their courts, configure pricing and timeslots, and interact with customers through dedicated management dashboards within a single scalable platform. Owners can register their futsal center, activate their listing through a subscription payment, add and manage pitches, handle booking requests, and access detailed analytics dashboards for business insights. User reviews are automatically processed using the TF-IDF feature extraction algorithm and classified by the Logistic Regression model to determine positive or negative sentiment, providing meaningful insights for both players and futsal owners. The Haversine formula is applied to calculate real-world distances between users and futsal centers, enabling the platform to recommend the nearest and most relevant courts based on the user's current location. A dedicated forum module allows users and futsal owners to engage in discussions, share experiences, arrange matches, and build a collaborative sports community around the platform. Administrators have full control over the platform, managing all tenants, user accounts,

subscription plans, forum content, and system-wide operations through a centralized admin panel.

Throughout the development process, the Agile iterative methodology was followed, ensuring continuous improvement, rapid feature delivery, and adaptability to feedback across all development sprints. Comprehensive testing including unit testing, system testing, and security testing was conducted to ensure that the platform is functional, secure, and reliable for real-world deployment.

By implementing Futsal, players can discover and book futsal courts with ease, overcome scheduling challenges, and engage with a growing sports community. Futsal owners benefit from a professional digital presence, streamlined booking management, and data-driven business insights. Overall, the Futsal platform successfully addresses the identified gap in Nepal's futsal industry by delivering an efficient, intelligent, and scalable digital solution for futsal facility management.

6.2 Future Recommendations

There are several opportunities for future improvements to further enhance user experience, accessibility, platform capabilities, and business growth.

1. Mobile Application Development

Dedicated Android and iOS mobile applications will be developed in future updates to make futsal discovery and booking more accessible on the go. The mobile apps will also support push notifications for booking confirmations, slot reminders, owner alerts, and forum activity, ensuring that users and owners stay informed and engaged at all times directly from their mobile devices.

2. Advanced AI and ML Enhancements

The existing sentiment analysis model will be expanded by training it on larger and more diverse real-world futsal review datasets to improve overall classification accuracy. An AI-powered chatbot will be introduced to assist users with booking queries, court recommendations, and platform navigation in real time. Additionally, collaborative filtering-based recommendation algorithms that consider individual user booking history and

preferences will be implemented to deliver more personalized and relevant court suggestions to each user.

3. Tournament and Match Management

A dedicated tournament management module will be introduced to allow futsal owners to organize and host futsal tournaments directly through the Futsal platform. Users will be able to register teams, view tournament brackets, track match results, and manage player rosters within the platform, transforming Futsal from a simple booking platform into a comprehensive futsal community and sports management ecosystem.

By implementing these future enhancements, Futsal can evolve into a world-class sports facility management platform that offers a richer, more personalized, and more intelligent experience for players, futsal owners, and administrators across Nepal and beyond. These improvements will strengthen the platform's competitive position, increase user engagement, and ensure long-term scalability and sustainability of the Futsal ecosystem.

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Appendices

Appendix A: Sample Module Designs and Source Code

This appendix presents sample module designs and code snippets used in the development of the Futsal Booking and Management System.

A.1 User Registration and Authentication Module

The User Registration and Authentication Module was developed using Node.js, Express.js, JWT, and bcrypt to ensure secure user authentication and authorization.

Sample Code:

```
const bcrypt = require("bcryptjs");
const hashedPassword = await bcrypt.hash(
  password,
  parseInt(FOOTSAL_PASSWORD_SALT_ROUNDS)
);
const newFutsal = await Futsal.create({
  futsalName: futsalName,
  ownerName,
  email,
  password: hashedPassword,
  phoneNumber,
}); const otp = generateOTP(6);
const otpData = {
  email,
  otp,
  type: "futsal_registration"};
redisClient.setEx(
  `futsal:otp:${email}`,
  300,
  JSON.stringify(otpData)
);

//sample login

const isMatch = await bcryptjs.compare(
  password,
  user.password
);

if (!isMatch) {
  return res.status(400).json({
```

```

error: "Invalid email or password",
});
}

const usertoken = generateJwt({
  id: user.id,
  email: user.email,
  role: user.role},
JWT_SECRET_USER,
"30d");

res.cookie("utoken", usertoken, {
  httpOnly: true,
  secure: NODE_ENV === "production",
  sameSite: "lax",
});

```

Sample Design



REGISTER YOUR FUTSAL

Enter your futsal and owner details.

Futsal Name *

Owner Name *

Owner Email *

Phone Number *

Continue



SET YOUR CREDENTIALS

Set up your futsal account credentials.

[← Back](#)

Futsal Account Email *

Password *

Confirm Password *

→] Complete Registration

Appendix A.2: Futsal Management Module

The Futsal Management Module enables futsal owners to efficiently manage futsal center information including futsal name, location, available pitches, pricing details, and booking time slots

Sample Code:

```
router.get('/futsals-profile', isFutsalAuthenticated, getFutsalProfile);

router.patch('/futsals-profile', isFutsalAuthenticated, updateFutsalProfile);

router.get('/futsal/:futsalId/pitches', getPitches);

router.get('/futsal/:futsalId/pitches/:id', resolveFutsalTenantWithToken, getPitchById);

router.post('/futsal/pitches/create', isFutsalAuthenticated, createPitch);

router.put('/futsal/pitches/edit/:id', isFutsalAuthenticated, editPitch);

router.get('/futsal/:futsalId/timeslots', getTimeslot);

router.post('/futsal/timeslots/create', isFutsalAuthenticated, createTimeslot);

router.put('/futsal/timeslots/update/:id', isFutsalAuthenticated, updateTimeslot);

router.delete('/futsal/timeslots/delete/:id', isFutsalAuthenticated, deleteTimeslot);

router.get('/futsal-bookings', isFutsalAuthenticated, getBookingsByAdmin);

router.post('/futsal/bookings/offline', isFutsalAuthenticated, createBookingByAdmin);

router.get('/futsal/bookings/stats', isFutsalAuthenticated, getBookingStats);

router.get('/futsal/bookings/:bookingId', isFutsalAuthenticated, getBookingByIdAdmin);

router.patch('/futsal/bookings/:bookingId/cancel', isFutsalAuthenticated,
cancelBookingByAdmin);

router.patch('/futsal/bookings/:bookingId/confirm', isFutsalAuthenticated,
confirmBookingByAdmin);

router.patch('/futsal/bookings/:bookingId/reject', isFutsalAuthenticated,
rejectBookingByAdmin);

router.patch('/futsal/bookings/:bookingId/unconfirm', isFutsalAuthenticated,
unconfirmBookingByAdmin);
```

```
router.delete('/futsal/bookings/:bookingId', isFutsalAuthenticated, deleteBookingByAdmin);

router.post('/futsal/media/upload', isFutsalAuthenticated, mediaUpload.array("media"),
uploadMedia);

router.post('/futsal/media/pitch/:pitchId/upload', isFutsalAuthenticated,
mediaUpload.array("media"), uploadPitchMedia);

router.post('/futsal/media/pitch/:pitchId/facility-upload', isFutsalAuthenticated,
mediaUpload.array("media"), uploadFacilitiesMediaByPitchId);

router.delete('/futsal/media/category/:category', isFutsalAuthenticated,
deleteMediaByCategory);

router.delete('/futsal/media/category/:category/pitch/:pitchId', isFutsalAuthenticated,
deleteMediaByCategory);

router.delete('/futsal/media/:id', isFutsalAuthenticated, deleteMediaById);

router.post('/futsal/info/create', isFutsalAuthenticated, createInfo);

router.get('/futsal/:futsalId/info/', getInfo);

router.put('/futsal/info/:infoId', isFutsalAuthenticated, updateInfo);

router.get('/futsal-location', isFutsalAuthenticated, getFutsalLocation);

router.post('/futsal/location/create', isFutsalAuthenticated, postFutsalLocation);

router.put('/futsal/location/edit/:locationId', isFutsalAuthenticated, editFutsalLocation);

router.get('/futsal/:futsalId/location/', getFutsalLocationByUser);

router.get('/futsal-ratings', isFutsalAuthenticated, getRatings);

router.get('/futsal/:futsalId/ratings', getRatingByfutsalId);

router.delete('/futsal/ratings/:ratingId', isFutsalAuthenticated, deleteRatingByAdmin);

router.get('/futsal-visitors', isFutsalAuthenticated, getVisitorsDetails);

router.post('/futsal/:futsalId/visitors/track', optionalUserAuth, trackVisitor);

router.get('/futsal/contact', isFutsalAuthenticated, getContacts);
```

Sample Design :

Location (Mandatory)

DISTRICT

kathmandu

CITY

manamajju

STREET ADDRESS

manamajju pool

POSTAL CODE

44600

Click on the map to place the pin, or use your current location. The selected coordinates are saved automatically

Latitude: 23.787878

Longitude: 72.23564

Use My Current Location

tarkeshor municipality 12

Save Location

Facility Info (Mandatory)

ESTABLISHED YEAR

2010

WEBSITE LINK

https://allfutsal.com

FACILITIES

Add facility (e.g. Parking)

drinking water

bathroom

bike parking

OPERATING HOURS

Monday

06:00 AM

10:00 PM

☐ Closed

Tuesday

06:00 AM

10:00 PM

☐ Closed

Wednesday

06:00 AM

10:00 PM

☐ Closed

Thursday

06:00 AM

10:00 PM

☐ Closed

Friday

06:00 AM

10:00 PM

☐ Closed

Saturday

06:00 AM

10:00 PM

☐ Closed

Sunday

06:00 AM

10:00 PM

☐ Closed

FACEBOOK URL

https://facebook.com

INSTAGRAM URL

https://instagram.com

YOUTUBE URL

https://youtube.com/@yourchannel

X / TWITTER URL

https://x.com/yourpage

TIKTOK URL

https://tiktok.com

car are not allowed

Pitch Management

Add, configure, and maintain your Futsal pitches.

+ Add Pitch

Total Pitches

3

Active Pitches

2

Maintenance

0

Today's Revenue

Rs 0

pitch A

PRICE / HR

Rs 1000.00

DAILY REVENUE

Rs 0

SURFACE TYPE

green

DIMENSIONS

10*18 dm

TODAY'S BOOKINGS

0

Edit

Schedule

pitch B

PRICE / HR

Rs 1200.00

DAILY REVENUE

Rs 0

SURFACE TYPE

green

DIMENSIONS

30m x 20m

TODAY'S BOOKINGS

0

Edit

Schedule

Pitch C

PRICE / HR

Rs 1100.00

DAILY REVENUE

Rs 0

SURFACE TYPE

green

DIMENSIONS

30m x 20m

TODAY'S BOOKINGS

0

Edit

Schedule

TODAY'S REVENUE

Rs 0

+Rs 0 FROM YESTERDAY

THIS WEEK

Rs 3000

+200% FROM LAST WEEK

THIS MONTH

Rs 5,000

+Rs 5000 FROM LAST MONTH

THIS YEAR

Rs 5,000

+0% FROM LAST YEAR

Revenue Trend

Last 7 days

FRI

SAT

SUN

MON

TUE

WED

THU

Recent Transactions

Aashis Rijal

pitch A • 08:00:00

Rs 1000

CARD

anish rijal

pitch A • 10:10:00

Rs 1000

CARD

bak

pitch A • 09:30:00

Rs 1000

CARD

Aashis Rijal

pitch A • 07:30:00

Rs 1000

CARD

Bibek Adhikari

pitch A • 10:00:00

Rs 1000

CARD

lxxxii

Appendix A.3: Booking Management Module

The Booking Management Module is responsible for handling the core functionality of the system, allowing users and futsal owners to create, manage, and control futsal bookings efficiently.

Sample Code:

```
const router = express.Router();
// Admin (Futsal Owner) APIs
router.get("/futsal-bookings", isFutsalAuthenticated, getBookingsByAdmin);
router.post("/futsal/bookings/offline", isFutsalAuthenticated, createBookingByAdmin);
router.get("/futsal/bookings/stats", isFutsalAuthenticated, getBookingStats);
router.get("/futsal/bookings/:bookingId", isFutsalAuthenticated, getBookingByIdAdmin);
router.patch("/futsal/bookings/:bookingId/cancel", isFutsalAuthenticated,
cancelBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/confirm", isFutsalAuthenticated,
confirmBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/reject", isFutsalAuthenticated,
rejectBookingByAdmin);
router.patch("/futsal/bookings/:bookingId/unconfirm", isFutsalAuthenticated,
unconfirmBookingByAdmin);
router.delete("/futsal/bookings/:bookingId", isFutsalAuthenticated, deleteBookingByAdmin);

// User APIs
router.get("/futsal/:futsalId/bookings", isUserAuthenticated, getBookingsByUser);
router.post("/futsal/:futsalId/bookings", isUserAuthenticated, createBooking);
router.patch("/futsal/:futsalId/bookings/:bookingId/cancel", isUserAuthenticated,
cancelBooking);
router.delete("/futsal/:futsalId/bookings/:bookingId", isUserAuthenticated,
deleteBookingByUser);
module.exports = router;
```

Sample Design:

Booking Management

All bookings overview — newest first

+ Add Walk-in Booking

mm/dd/yyyy

Total

6

Confirmed

4

Pending

1

Revenue

Rs 5000

All Bookings

Every date - 6 total

All Statuses

CUSTOMER	DATE	PITCH	TIME SLOT	STATUS	PRICE	ACTIONS
Aashis Rijal 9847749977	2026-05-27	pitch A	08:00:00 - 09:00:00	CONFIRMED	Rs 1000.00	View Unconfirm Cancel
anish rijal 9800000001	2026-05-18	pitch A	10:10:00 - 11:10:00	PENDING	Rs 1000.00	View PAST SLOT
bbk 9800000000	2026-05-18	pitch A	09:00:00 - 10:00:00	CONFIRMED	Rs 1000.00	View PAST SLOT
Aashis Rijal 9841824710	2026-05-18	pitch A	07:00:00 - 08:00:00	CANCELLED	Rs 1000.00	View PAST SLOT
Bibek Adhikari 9849137565	2026-05-11	pitch A	10:00:00 - 11:00:00	CONFIRMED	Rs 1000.00	View PAST SLOT
Aashis Rijal 9841824710	2026-04-29	pitch A	21:01:00 - 22:01:00	CONFIRMED	Rs 1000.00	View PAST SLOT

Add Phone / Walk-in Booking

For customers who book by phone or in person

PHONE *

9841824710

CUSTOMER NAME

kedar Dhungana

PITCH *

pitch A

DATE *

05/26/2026

TIMESLOT *

10:00:00 - 11:00:00

AMOUNT (RS) *

1000.00

STATUS

Completed

DESCRIPTION / NOTES

cash remain

Cancel

Create Booking

Booking #22

Wednesday, 27 May 2026

CONFIRMED

Website / App

Booked through the website or app

CUSTOMER

NAME

Aashis Rijal

PHONE

9847749977

EMAIL

aashishrijal252@gmail.com

PITCH & SLOT

PITCH

pitch A

TYPE

SA · green

DIMENSIONS

10*10 dim

TIME

08:00 - 09:00 (Thu)

RATE / HR

Rs 1000.00

BOOKING

BOOKING ID

#22

DATE

Wednesday, 27 May 2026

AMOUNT

Rs 1000.00

CREATED

5/20/2026, 10:23:03 PM

NOTES

No additional notes

PAYMENT

STATUS

pending

GATEWAY

esewa

AMOUNT

Rs 1000.00

ORDER ID

93dc39d0-b755-45ec-92c2-741ee118d0f0

PAID AT

Not verified yet

Mark pending

Cancel booking

Appendix A.4: Review and Sentiment Analysis Module

The Review and Sentiment Analysis Module allows users to submit reviews and ratings for futsal centers and automatically analyzes textual feedback using machine learning techniques.

Sample code:

```
# Load model and vectorizer at module level

project_root = Path(__file__).resolve().parents[2]

model_dir = project_root / "app" / "trained_model"

model_path = model_dir / "sentiment_model.joblib"

vectorizer_path = model_dir / "tfidf_vectorizer.joblib"


if not model_path.exists() or not vectorizer_path.exists():

    raise FileNotFoundError("Model or vectorizer not found. Train model first.")

model = joblib.load(model_path)

vectorizer = joblib.load(vectorizer_path)

print("Sentiment model and vectorizer loaded successfully")


def _clean_text(text: str) -> str:

    text = str(text).lower().strip()

    text = re.sub(r"https?://\S+|www\.\S+", " ", text)

    text = re.sub(r"^[a-z0-9\s]", " ", text)

    text = re.sub(r"\s+", " ", text).strip()

    return text
```

```

def predict(text: str) -> dict:

    """Predict sentiment for given text."""

    if not text or not text.strip():

        return { "sentiment": "Unknown", "confidence": 0.0 }

    # Preprocess text

    cleaned_text = _clean_text(text)

    # Vectorize

    vectorized_text = vectorizer.transform([cleaned_text])

    # Predict

    prediction = model.predict(vectorized_text)[0]

    probabilities = model.predict_proba(vectorized_text)[0]

    # Map label

    label_map = {0: "Negative", 1: "Positive" }

    sentiment = label_map.get(prediction, "Unknown")

    positive_class_index = 1 if len(probabilities) > 1 else 0

    confidence = float(probabilities[positive_class_index])

    return { "sentiment": sentiment, "confidence": round(confidence, 4) }

```

Sample Design



Edit Your Review
Share your experience with this facility

Your Rating

Good — 3 stars

Your Review (optional)

Save Changes

Cancel

YOUR REVIEW



YOUR REVIEW

A

Aashis Rijal
Apr 10, 2026

68% POSITIVE





 Edit

 Delete

Appendix A.5: Forum Module

The Forum Module provides an interactive platform where users and futsal owners can post questions, share experiences, and engage in discussions related to futsal services.

Sample Code:

```
const router = express.Router()

router.post('/forum/create', isBothAuthenticated, createForum);

router.get('/forums', getAllForums);

router.get('/forum/user0', isUserAuthenticated, getForumsByUserId);

router.get('/forum/futsal', isFutsalAuthenticated, getForumsByFutsalId);

router.get('/forum', getForumsByCategory);

router.get('/forum/:forumId', getForumById);

router.get('/forum/slug/:slug', getForumBySlug);

router.delete('/forum/:forumId',
isBothAuthenticated, deleteForum);

router.post('/forum/:forumId/reply/create',
isBothAuthenticated, createForumReply);

router.get('/forum/:forumId/reply',
isBothAuthenticated, getRepliesByForumId);

router.get('/forum-reply',
isBothAuthenticated, getRepliesByUserIdOrFutsalId);

router.delete('/forum/reply/:replyId',
isBothAuthenticated, deleteForumReply);
```

Start a new topic

Title

test

Category

General

Details

test

Post to community

Latest Topics

All

Announcement

General

Help

HELP

Searching Best Futsal

Started by

Aashis Rijal

0 views

3/26/2026

1 Likes

Unlike

ANNOUNCEMENT

Introduce A1 futsal

Started by

Aashis Rijal

0 views

3/26/2026

0 Likes

Unlike

GENERAL

search best 5A futsal in manmai

Started by

Aashis Rijal

0 views

3/31/2026

0 Likes

Unlike

HELP

I need a player to play futsal today 5-6

Started by

Aashis Rijal

0 views

3/31/2026

0 Likes

Unlike

GENERAL

test

Started by

Unknown User

0 views

3/31/2026

1 Likes

Unlike

HELP

3/26/2026, 9:21:45 AM

Delete post

Searching Best Futsal

Aashis Rijal

User

search best futsal inside manamaiju

1 Likes

Unlike

2 Replies

0 Views

Discussion (2)

Aashis Rijal

OP

3/31/2026

premiere futsal

1 Likes

Unlike

lxxxix

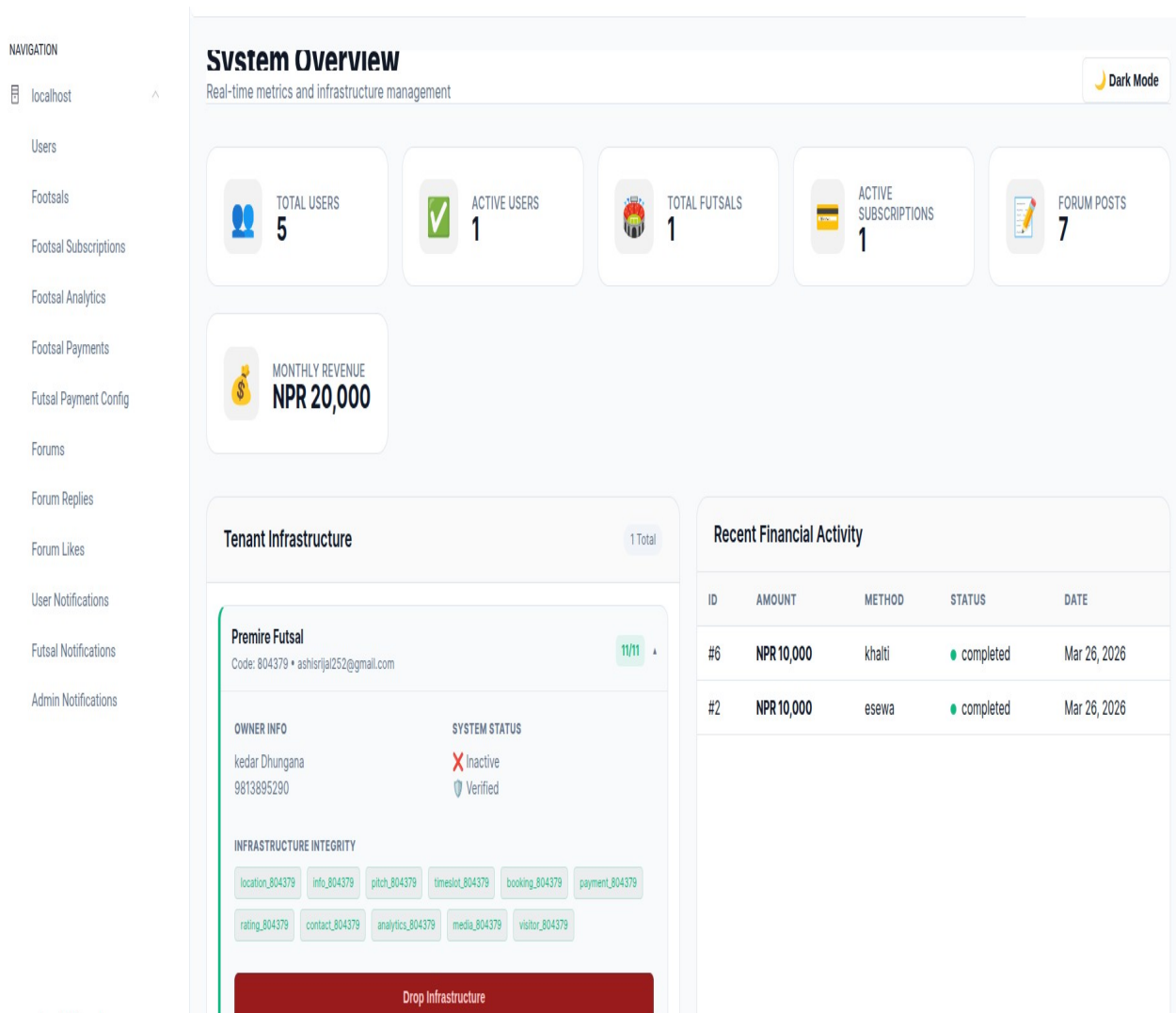
Appendix A.6: Admin Management Module

The Admin Management Module provides centralized administrative control over the futsal booking platform. It allows administrators to manage users, futsal owners, bookings, reports, forums, payments, and overall system activities.

Sample code:

```
// config/adminConfig.js
const AdminJS = require('adminjs');
const AdminJSExpress = require('@adminjs/express');
const AdminJSSequelize = require('@adminjs/sequelize');
const { sequelize, User, Futsal, Subscription, Payment } = require('../models');
const { ADMIN_COOKIE_SECRET, ADMIN_EMAIL, ADMIN_PASSWORD,
ADMIN_FRONTEND_URL } = process.env;
AdminJS.registerAdapter(AdminJSSequelize);
const setupAdminPanel = (app) => {
  const adminJs = new AdminJS({
    databases: [sequelize],
    rootPath: ADMIN_FRONTEND_URL || '/admin',
    branding: {
      companyName: 'NepFutsal Admin',
      softwareBrothers: false,
      logo: '/logo.png',
    }
  });

  const router = AdminJSExpress.buildAuthenticatedRouter(
    adminJs,
    {
      authenticate: async (email, password) => {
        if (email === ADMIN_EMAIL && password === ADMIN_PASSWORD) {
          return { email };
        }
        return null;
      },
      cookieName: 'adminjs',
      cookiePassword: ADMIN_COOKIE_SECRET,
    },
    null,
    { resave: false,
      saveUninitialized: true,
      secret: ADMIN_COOKIE_SECRET,
    }
  ); app.use(adminJs.options.rootPath, router);
  return adminJs;
};
module.exports = { setupAdminPanel };
```



Appendix A.7: Payment Management Module

The Payment Management Module manages online payment processing for futsal bookings using digital payment gateways such as Khalti and eSewa

Sample Code:

```
const initiateEsewaPayment = async (req, res) => {
  const { bookingId, amount } = req.body;
  const paymentData = {
    amt: amount,
    psc: 0,
    pdc: 0,
```

```

    txAmt: 0,
    tAmt: amount,
    pid: bookingId,
    scd: process.env.ESEWA_MERCHANT_CODE,
    su: `${process.env.BASE_URL}/payment/eseva/success`,
    fu: `${process.env.BASE_URL}/payment/eseva/failure`,
  }; const query = new URLSearchParams(paymentData).toString();
  const paymentUrl = `https://uat.eseva.com.np/epay/main?${query}`;
  return res.json({
    message: "eSewa payment initiated",
    paymentUrl,
  });
};const initiateKhaltiPayment = async (req, res) => {
  const { bookingId, amount, user } = req.body;

  const response = await axios.post(
    "https://a.khalti.com/api/v2/epayment/initiate/",
    {
      return_url: `${process.env.BASE_URL}/payment/khalti/success`,
      website_url: process.env.BASE_URL,
      amount: amount * 100, // Khalti uses paisa
      purchase_order_id: bookingId,
      purchase_order_name: "Futsal Booking",
      customer_info: {
        name: user.name,
        email: user.email,
        phone: user.phone,
      },
    },
  );
  {

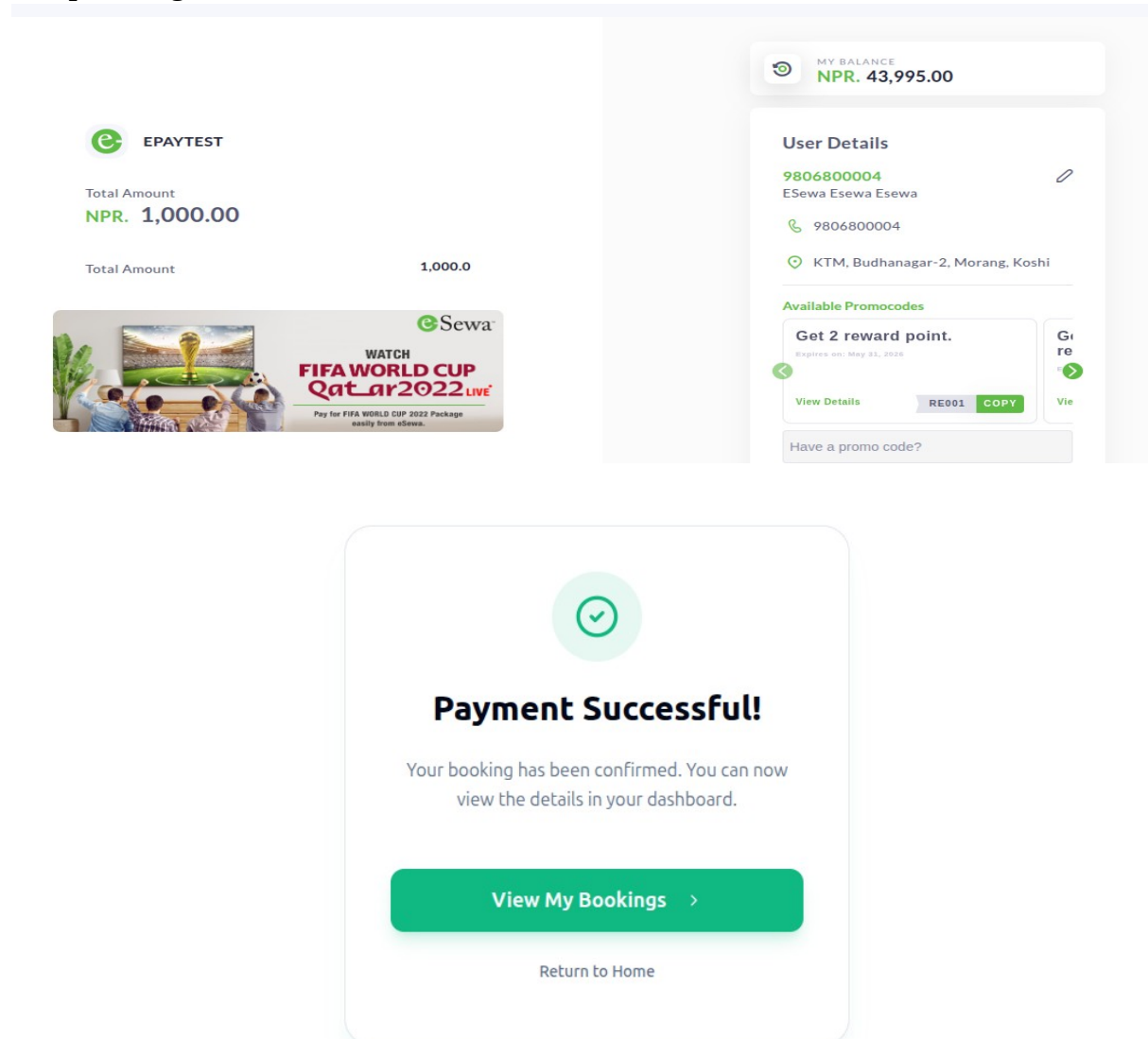
```

```

headers: {
  Authorization: `Key ${process.env.KHALTI_SECRET_KEY}`,
}, } );
return res.json({
  payment_url: response.data.payment_url,
});});

```

Sample Design:



Appendix B: Docker & Deployment Commands

This appendix includes common Docker and deployment commands used to containerize and run the futsal booking system in a production-like environment. Docker ensures consistency across development and deployment environments.

1. Build Docker Images

```
docker build -t futsal-booking-app .
```

2. Run Docker Container

```
docker run -p 3000:3000 futsal-booking-app
```

3. Run in Detached Mode

```
docker run -d -p 5000:5000 --name futsal-app futsal-booking-app
```

4. List Running Containers

```
docker ps
```

5. Stop Container

```
docker stop futsal-app
```

6. Remove Container

```
docker rm futsal-app
```

7. View Logs

```
docker logs futsal-app
```

8. Docker Compose up

```
docker-compsoe up
```

SUPERVISOR'S LOGS

2083/01130

Name	sig.
Nashish Rimal	
Dipesh Devkota	
Kedar Dhungana	
Bijaya Babu Bhatta	
Amisa Dhakal	
Adarsha Poudel	
Gharat Amgain	
Ashish Lamichhane	
Shubheshkshya Gautam	
Shrisha Poudel	

Group A	Signature
Nashish Rimal	
Kedar Dhungana	
Dipesh Devkota	
Group B	
Achal Shrestha	
Mura K.C.	
Roshan Shrestha	
Group C	
Roshan Bhakthari	
Sangam Dhital	

Date: 2082-12-5

Name	Signature	TOPIC DISCUSS
Amisa Dhakal		
Yunisha Poudel		
Bijaya Babu Bhatta	Absent	
Nashish Rimal		
Dipesh Devkota		
Kedar Dhungana	Absent	
Shubheshkshya Gautam		
Shrisha Poudel		
Adarsha Poudel	Absent	

Group Members		Signature
① Yunisha Poudel		<u>Yunisha</u>
② Amisa Dhokai		X
③ Bijaya Babu Bhatta		X
① Aashish Rijal		<u>Aashish</u>
② Kedar Dhungana		X
③ Dipesh Devkota		X
① Bharat		X
② Aashish Lamichhane		<u>Aashish</u>
③ Kushay Shrestha		X

SUPERVISOR EVIDENCE

Futsal Bookings and Management Platform

A project report submitted for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer science & Information Technology awarded by Tribhuvan University.

Submitted By

AASHISH RIJAL
DIPESH DEVKOTA
KEDAR DHUNGANA

Submitted To

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Gongabu, Kathmandu Nepal



Date: 3rd Chaitra, 2082